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Lumbar disc herniations are a common manifestation of degenerative disease.¹⁻³ They tend to occur early within the degenerative cascade, representing the tensile failure of the annulus to contain the gel-like nuclear portion of the disc. With improvements in advanced imaging techniques, lumbar disc herniations have been increasingly recognized in symptomatic and asymptomatic individuals.⁴

Treatment decision making for patients with herniated discs can be challenging. Nonoperative treatment can be effective in most cases.⁵⁻⁹ Other authors have indicated that surgery leads to superior results, especially in short-term pain relief.^{1,7-10} Several authors have highlighted the influence of fragment location and pattern, as well as social and psychological factors on outcomes.^{7-9,11-13} The exact natural history and complex interaction of biologic, psychosocial, ergonomic, and cultural variables have not been well established.

In the best-case scenario, the clinician can radiologically identify a single culprit disc that positively correlates with clinical findings. In patients who do not respond to nonoperative management, disagreement remains concerning the optimal period of observation, timing of surgery, method of excision, and type of postoperative rehabilitation. In less evident cases, one or more minor disc bulges may be identified that are difficult to attribute to the patient's signs and symptoms. The use of diagnostic injections can be helpful in localizing symptomatic regions. Empiricism and reliance primarily on intuition may lead to inferior results. Strict agreement between a patient's signs, symptoms, and correlative diagnostic tests needs to exist when predicating treatment recommendations.

To address these questions, the wealth of classic and contemporary contributions made to the understanding of lumbar disc herniations has been reviewed. This chapter synthesizes the information and organizes it to help clinicians' understanding and recommendations of management of this seemingly simple, but realistically challenging, problem.

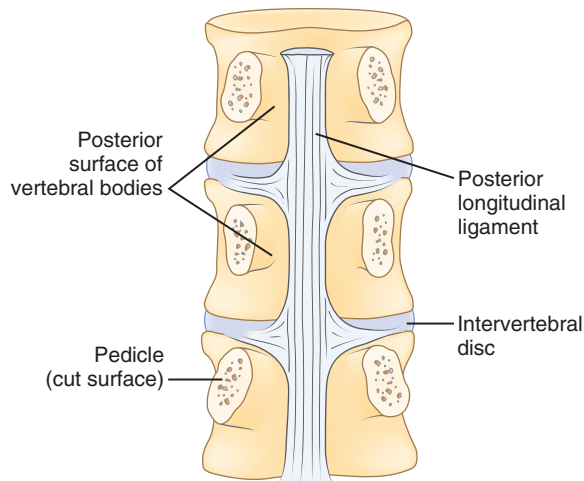
Pathoanatomy

Effective evaluation is based on an intimate understanding of the relationship of the lumbar intervertebral disc to its

surrounding structures. The disc is the anterior border of the spinal canal at the facet joint level. It is covered by the thin posterior longitudinal ligament, which is concentrated in the midline, from which small bands extend laterally to cover the inferior aspect of the disc (Fig. 47.1). This configuration leaves the superior part of the posterolateral disc bare, and is thought to contribute to the fact that posterolateral (or paracentral) herniations are the most frequent location for herniations to occur. Cumulative degenerative changes occur in this region of the disc from concentration of torsional, axial loading, and flexion-induced biomechanical strains.

The spinal cord ends at approximately the L1 level in adults to form the conus medullaris. The cauda equina is located within the lumbar spinal canal. It contains the lumbar and sacral nerve roots bathed in cerebrospinal fluid contained— or encapsulated—by the pia, arachnoid, and dural membranes (meninges). Nerve roots branch from the cauda equina one level above their exiting foramen (Fig. 47.2). The L5 nerve root leaves the cauda equina approximately at the level of the L4 vertebral body. It descends inferolaterally to pass anterior to the L4–L5 facet joint and posterior to the L4–L5 disc. Intimately associated with the inferomedial aspect of the L5 pedicle, the root turns lateral to enter the L5–S1 intervertebral (neural) foramen just proximal to the L5–S1 disc. Within the foramen, sensory cell bodies form the dorsal root ganglion. The root, now called a *postganglionic spinal nerve*, exits the neural foramen, where it is in close proximity to the lateral aspect of the L5–S1 disc. Fibrous bands (called *Hoffman ligaments*) often tether the nerve to the disc in this region.^{14,15} After a short extraspinal course, the nerve divides into a ventral and dorsal primary ramus.

The location of the disc herniation determines which root is primarily affected. The spinal canal can be divided into longitudinal zones (Fig. 47.3). The *central zone* is delineated by the lateral borders of the cauda equina. The *lateral recess* is between the lateral border of the cauda equina and the medial border of the pedicle. Although the term *lateral recess* is frequently used to describe stenosis from bony encroachment (*lateral recess stenosis*), it sufficiently describes the location of *paracentral*, *posterolateral*, or *juxtacentral* herniations. Within the lateral recess, fragments medial to the nerve root,



**Anterior vertebral segments
posterior view
(pedicles sectioned)**

FIG. 47.1 The posterior longitudinal ligament incompletely covers the posterior portion of the disc. Specifically, the superolateral aspect of the disc remains uncovered, which may help explain why disc herniations are most common in this region.

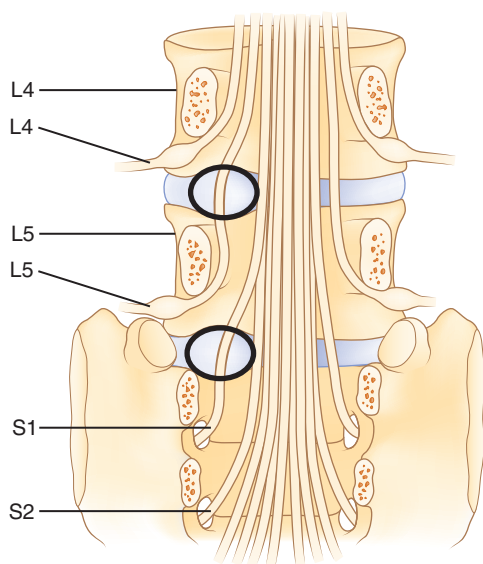


FIG. 47.2 Lumbar nerve root branches exit the dural sac one vertebral level above their respective foramen. Paracentral disc herniations tend to affect the traversing nerve root as it crosses the intervertebral disc.

interposed between it and the cauda equina, are called *axillary* herniations (Fig. 47.4). The *foraminal* zone is between the medial and lateral borders of the pedicle. Herniations beyond the lateral border of the pedicle are within the far-lateral or *extraforaminal* zone. Herniations in the foraminal or extraforaminal zones usually affect the exiting nerve.

Fragments can displace cranially or caudally. Axillary herniations have a tendency to migrate distally, lying inferior to the disc space. Superior migration of the fragment can position it behind the adjacent cranial vertebral body. Locating the fragment preoperatively is crucial to successful operative excision.

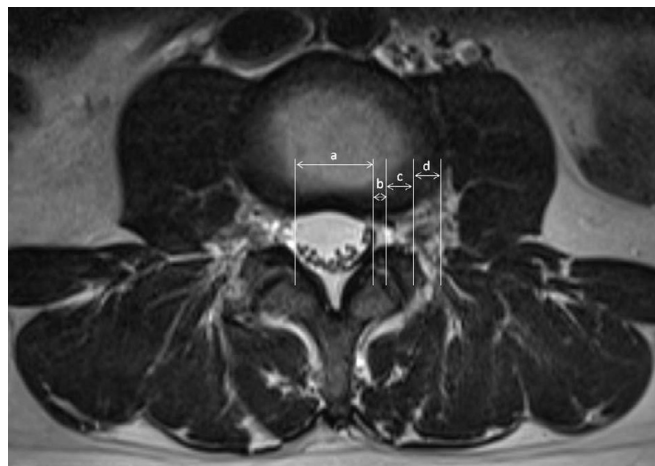


FIG. 47.3 Considered in cross-section, the spinal canal can be divided into anatomic zones to describe better the location of lumbar disc herniations. The *central zone* (a) is within the borders of the cauda equina dural sac. The *lateral recess* (b), which paracentral disc herniations can compromise, is bordered by the lateral aspect of the dural sac and the medial aspect of the pedicle and neural foramen. In this zone, the nerve root descends within the spinal canal toward its respective foramen. The *foraminal zone* (c) is the space between adjacent ipsilateral pedicles. The *extraforaminal zone* (d) is the space lateral to the pedicles. Disc herniations in this region, commonly referred to as *far lateral*, affect the exiting nerve root.

Pathophysiology

Disc Degeneration and Herniation

Disc herniation is one stage of the lumbar degenerative cascade. It is considered one of the earlier stages, following internal disc disruption. Herniation occurs through a tear in the annulus fibrosus. The annulus is the thick outer layer that normally withstands tensile forces transferred from the compressed nucleus pulposus (Fig. 47.5).^{16,17} Force transfer works only if the nucleus-annulus-endplate complex acts as a closed-volume system.¹⁸ Normally, compression across the disc space leads to increased pressure within the nucleus. The soft nucleus deforms and flattens, pushing against the annular fibers, which then generates tensile hoop stresses. The circumferential fibers are placed under tension, dissipating stresses and containing the annulus.

With disruption of the annulus, the soft nucleus can be pushed through (i.e., herniated) if placed under sufficient pressure. The nucleus must be fluid, or “dynamic,” enough to permit herniation to occur. Discs in younger individuals that have a well-hydrated nucleus are more likely to herniate. Older patients with desiccated discs are less prone to herniation. The ejected portion is typically a fibrocartilaginous fragment.¹⁹ In some cases, a piece of annulus or endplate fibrocartilage can be associated with it. In juveniles, an apparent herniation may represent a Salter type II fracture of the vertebral ring apophysis with its attached annulus.

When a portion of the nucleus is ejected, disc mechanics are altered. Frei and colleagues¹⁷ showed that nucleotomy alters the loading pattern across the disc space, with the annulus sustaining higher compression forces than normal.

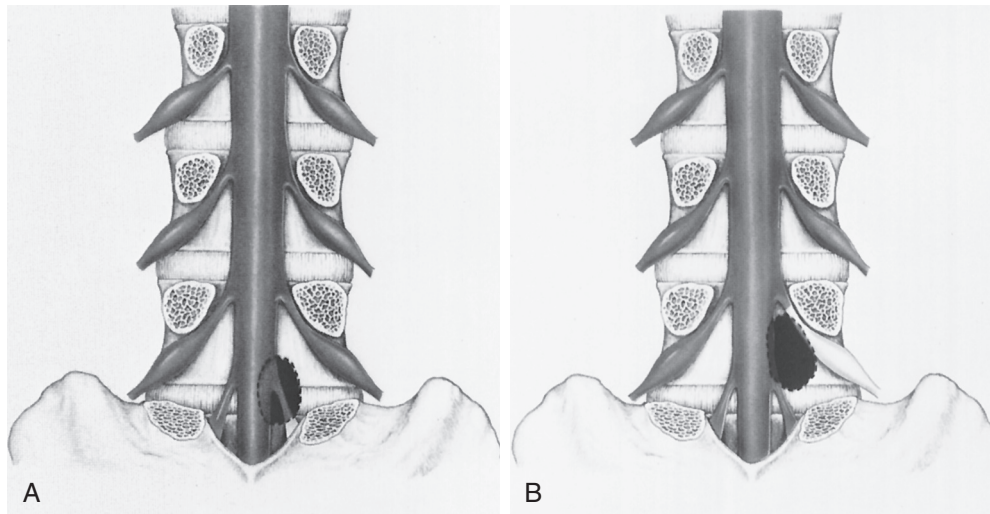


FIG. 47.4 (A) Most commonly, paracentral disc herniations compress the traversing (descending) nerve root along its lateral aspect. (B) In some cases, the disc fragment can be interposed between the nerve root and lateral border of the cauda equina sac. These are known as *axillary disc herniations*.

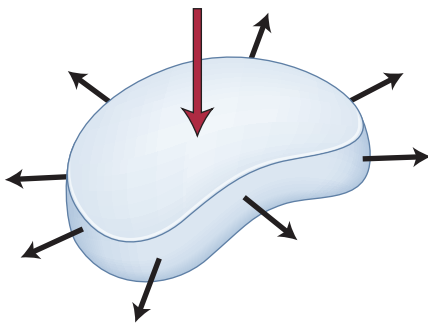


FIG. 47.5 The thick, outer annulus normally withstands tensile forces transferred from a compressed nucleus pulposus. This force transfer works, however, only if the nucleus-annulus-endplate acts as a closed-volume system. An annular defect disrupts this closed-volume system and can allow a portion of the nucleus to escape (or herniate).

This situation can lead to increases in endplate pressures along the periphery where the annulus attaches to the bone. Chondro-osseous metaplastic changes such as osteophytes or sclerosis in these regions are a response to long-standing abnormal loading patterns.

The exact inciting event leading to disc herniation is unknown. Some authors believe that an acute traumatic episode leads to displacement of the disc, although this is most likely related to force imparted onto a previously degenerated disc, which has developed a focal annular weakness. Acute sciatica from a disc herniation is often associated with a prodromal history of back pain.

Postural variations can influence intradiscal pressures. The highest pressures have been recorded in patients with the torso forward flexed with weight in hand. In an elegant biomechanical study, Wilder and colleagues¹⁶ found that combined lateral bend, flexion, and axial rotation with 15 minutes of exposure to vibration can lead to tears extending from the nucleus across the annulus. This finding may have significance for occupations with exposure to long periods of vibratory stimuli, such as truck drivers and machine workers.

Disc Herniation and Sciatica

The most classic symptom of a herniated disc is radicular pain in the lower extremity following a dermatomal distribution. Focal neurologic deficits attributable to the same nerve root are sometimes present and lend further diagnostic accuracy. The relationship between disc herniation and sciatica is incompletely understood, however.

In animals and humans, pure compression of a noninflamed nerve produces sensory and motor changes without pain, whereas pain is elicited with manipulation of inflamed nerves.²⁰ These findings suggest that herniated discs large enough to cause mechanical compression of a nerve root may produce focal deficits, but that associated sciatic-type pain is produced only if the nerve root is concurrently irritated or inflamed. Inflammation may be produced by prolonged neuroischemia of the microvasculature of the nerve root from mechanical compression or by nonmechanical, possibly biochemical, factors. This phenomenon helps explain why some patients with small bulges or protrusions contacting inflamed nerves have pain that does not seem to be consistent with the “small” degree of neural compression. Additionally, these patients frequently do not have demonstrable sensory or motor deficit.

Neurochemical factors also have a role in the production of sciatic pain. This role may be related to initiation of an immune response locally or systemically, or both. Spiliopoulou and colleagues²¹ examined immunoglobulin G (IgG) and IgM levels in discs excised from patients with sciatica and controls. Although IgG levels were equivalent, elevated levels of IgM were found in discs from sciatica patients but not in controls, suggesting a local and humoral antigenic inflammatory reaction as a contributor to pain. Other investigators have shown the role of cytokines in the mediation of root pain. Olmarker and Rydevik²² studied the effects of selective inhibition of tumor necrosis factor- α (TNF- α) in a herniated disc model in pigs. They found preservation of nerve conduction velocity and decreased nerve root injury in treated animals versus

controls, suggesting a role of TNF- α in potentiating nerve dysfunction.

Similarly, research has suggested that matrix metalloproteinase, nitric oxide, prostaglandin E₂, and interleukin-6 in discs excised from patients with herniation and radiculopathy may have a causative role in pain production.²³ A more recent investigation was unable to confirm the presence of these inflammatory markers in the epidural space of patients with symptomatic disc herniations, however.²⁴ Other investigators have shown that in extruded or sequestered discs, a cellular inflammatory reaction may be locally mediated via T cells and macrophages²⁵; this has been postulated to play a role in herniated disc regression.²⁶

There is evidence of systemic inflammatory responses to disc herniations as well. Brisby and colleagues²⁷ detected elevated levels of glycosphingolipid antibodies in the serum of patients with sciatica and disc herniation compared with healthy volunteers. Elevations were equivalent to those found in patients with autoimmune neurologic disorders, such as Guillain-Barré syndrome. Brisby and colleagues²⁷ suggested that a systemic autoimmune response to disc tissue may result in damage, or alteration, of nerve tissue. After age 12 years, the endplate apophyseal vessels close, which may facilitate an amnestic antigenic response to exposure to extruded nucleus pulposus tissue. These findings are helpful in considering patients who have severe sciatic pain with minimal mechanical compression and patients who seem to have persistent symptoms despite surgical decompression.

Disc Herniation and Back Pain

Most patients with symptomatic disc herniations present with leg and back pain (Table 47.1). The mechanism of degenerative back pain remains elusive, although many authors have suggested mechanisms. Accepting that herniation is a stage within the continuum of lumbar degeneration, discogenic pain generators may be a factor. Innervation of the posterior anulus by branches of the sinuvertebral nerve have been well documented and are a suggested pathway of nociceptive pain transmission from disc degeneration. An annular tear and nuclear herniation could result in similar pain transmission.

The concept of *vertebrogenic pain* has also been suggested. Jinkins and colleagues²⁸ studied the contribution of anterior disc herniations to back pain. They believed that the pain was

neurally mediated through branches of the ventral ramus and paravertebral autonomic plexus. Because the herniations were outside the spinal canal, they were not associated with compression of the cauda equina or nerve roots, but most patients complained of lower extremity paresthesias, mostly bilateral, in addition to low back pain. A direct causal relationship between anterior disc herniations and leg symptoms has not been clarified.

Classification of Disc Herniations

Classification of any disorder should be based on identifiable features that have some influence on prognosis or treatment decision making. Many classification systems have been proposed for lumbar disc herniations, although none are all-inclusive or ideal.^{29,30} It is more appropriate to consider them as tools to describe the herniation.

Morphology

Disc herniations can be described by their morphology. Before the introduction of advanced imaging, morphology was difficult to assess preoperatively. Currently, and to a lesser extent, computed tomography (CT) can differentiate disc morphology with reasonable reliability. Spengler and colleagues¹³ divided herniations into three types (Fig. 47.6). A *protruded* disc was defined as eccentric bulging through an intact anulus fibrosus. An *extrusion* was defined as disc material that crosses the anulus but is in continuity with the remaining nucleus within the disc space. A *sequestered* disc represents a herniation that is not continuous with the disc space; this is the typical “free fragment.”

Other authors have classified discs as either contained or uncontained.³¹ *Contained* disc herniations are subligamentous. It is presumed that they have not passed beyond the limits of the posterior longitudinal ligament or the outer layer of the anulus. *Uncontained* disc herniations have crossed this boundary. Advocates of this system describe contained and uncontained extrusions, with the former remaining beneath the outer layers of the anulus.³¹

Location

Herniations can be described topographically according to anatomic location (see Fig. 47.3). The herniation can be located within the central zone, lateral recess, foraminal, or extraforaminal regions. Herniations can also exhibit cranial or caudal migration in relation to the disc space.

Timing

Lumbar disc herniations can be organized according to the time from initial symptom onset. These may be arbitrarily divided as acute or chronic. Acute herniations are present for less than 3 to 6 months, whereas chronic discs cause symptoms for a longer time. Breakdown according to this time frame is based on our sense of what is a reasonable cutoff point. Because

TABLE 47.1 Prevalence of Back Pain and Sciatica in Adults	
Characteristic	Prevalence (%)
Any low back pain	60–80
Any low back pain persisting at least 2 wk	14
Low back pain persisting at least 2 wk at a given time (point prevalence)	7
Back pain with features of sciatica lasting at least 2 wk	1.6
Lumbar spine surgery	1–2

From Deyo RA, Loeser J, Bigos S. Herniated lumbar intervertebral disc. *Ann Intern Med.* 1990;112:598–603.

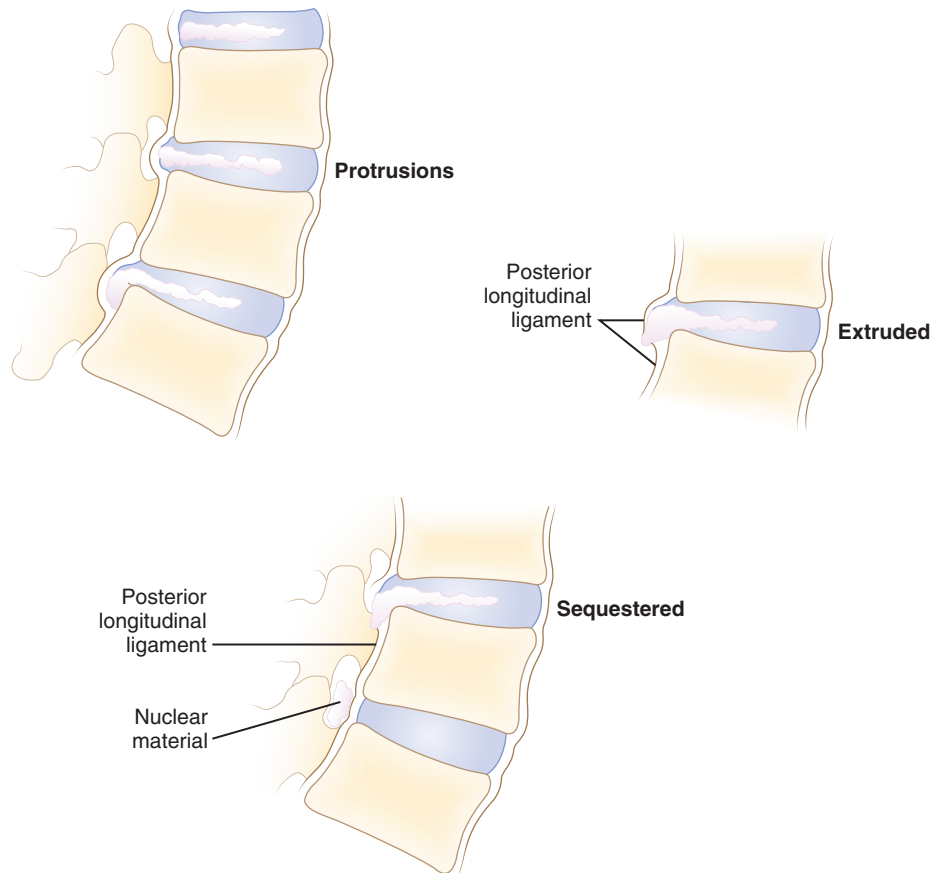


FIG. 47.6 Classification of disc herniations as described by Spengler and colleagues.¹³ *Disc protrusion* is defined as a bulging, displaced nucleus that has not extended beyond the limits of the annulus fibrosus. *Extrusion* extends beyond the annulus fibrosus but is still in continuity, at least partially, with the parent disc. *Sequestered* disc herniation implies that a fragment has broken free (i.e., free fragment) and is no longer in continuity with the parent disc. In some cases, in which disc herniation lies immediately behind the vertebral body, it is difficult to tell from which disc the herniated fragment originated.

the results of disc excision seem to be influenced by the timing of surgery, this categorization is important. From a survey of the literature, it seems that the results of disc excision are compromised if delayed more than 2 to 16 months from symptom onset.^{12,32–34}

History and Symptoms

Many patients describe a prodromal history of long-standing mild to moderate back pain. Although trauma is not the only component leading to a disc herniation, some patients describe a specific incident attributable to the onset of leg and back pain. This incident may be a fall, a twist, or lifting of a heavy item. Specific postures can lead to exponential increases in intradiscal pressure, which can predispose to disc injury.¹⁸ Exposure to vibrational energy combined with sustained lateral flexion and rotation may also predispose to herniation.¹⁶ The exact history of the incident and the presence of preexistent back or leg pain must be explored; this is particularly important in work-related injuries.

Pain is the most common complaint. Axial back pain is typically present, although some patients do not have this

complaint. Radicular pain is more typical and often the more “treatable” of the complaints. The pattern of lower extremity radiation depends on the level of the herniation. Lower lumbar or lumbosacral disc herniations can lead to the classic symptoms of pain radiating below the knee. Often pain extends into the foot and can follow a dermatomal distribution. S1 radicular pain may radiate to the back of the calf or the lateral aspect or sole of the foot. L5 radicular pain can lead to symptoms on the dorsum of the foot (Fig. 47.7). Radiculopathy from involvement of the upper lumbar roots can lead to more proximal symptoms. L2 and L3 radiculopathy can produce anterior or medial thigh and groin pain. Groin pain may also be indicative of L1 pathology. Radicular pain can be difficult to discern and is often not “classic.” Many patients do not exhibit pain in a specific dermatomal distribution, or the radiation does not extend along the entire leg. It may radiate only into the hip region or just the foot or any portion of the leg.

The character of radicular pain can be sharp, dull, burning, or dysesthetic. It can be exacerbated by coughing, bending, or lifting. A relieving maneuver may be lying supine with the knees and hips flexed, particularly with lower lumbar herniations. In contrast to patients with lumbar stenosis, patients

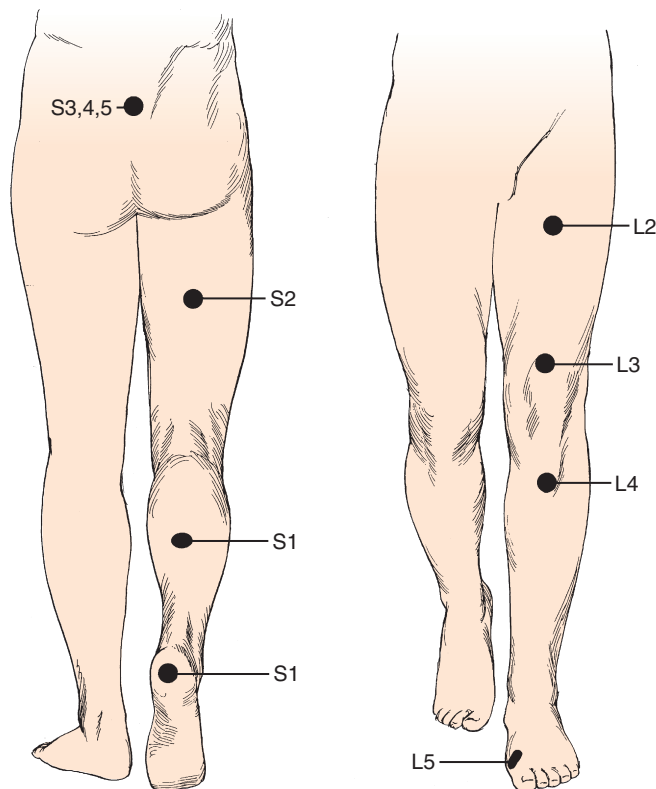


FIG. 47.7 The location of the pain can help localize the nerve root involved. Pain may radiate to small, isolated areas along the course of the dermatome.

with disc herniations more typically complain of constant pain that is not exacerbated by ambulation. Buttock pain is also common and can be referred or radicular in nature. Patients may interpret this as back pain, but the pathoanatomic significance of pain in this area is different than true axial pain. Pain is one component of radiculopathic complaints. Nerve compression can lead to motor and sensory deficits. Weakness may be reported as a slapping gait, footdrop, knee buckling, or imbalance when walking.

It is important to ask questions pertaining to bowel and bladder function. The examiner should inquire about urgency and frequency and fecal and urinary incontinence. Questions must be direct because most patients report constipation, which is often related to narcotic usage or inactivity, as a change in bowel habits. Acute bowel or bladder symptoms can be the sign of a cauda equina syndrome, which should mandate urgent surgical decompression.

The influence of social and psychological factors on the outcomes of disc surgery has been well documented. It is highly recommended to obtain a social and at least cursory psychiatric history. Prescription use of antidepressants is an important clue, although depression is often undiagnosed and untreated at the time of initial presentation. Other personality factors—such as chronic headaches, hysteria, hypochondriasis, nervous disorders, and impulsivity—can also be influential.^{13,35} Work history, pending litigation, and type of work should be obtained. A history of smoking is an independent

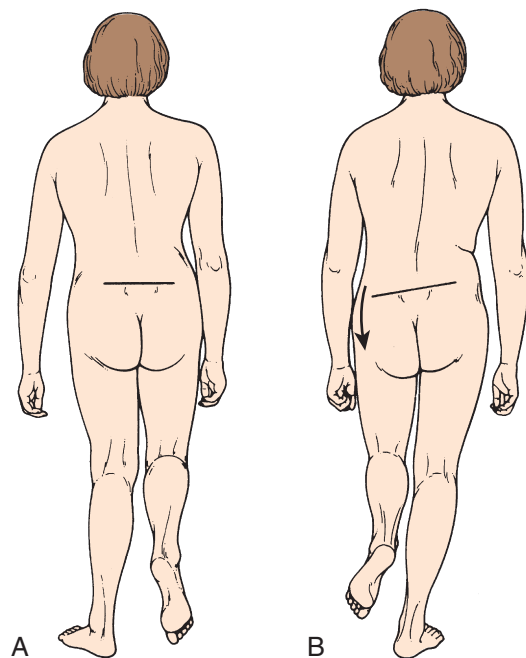


FIG. 47.8 (A) With normal L5 nerve root function, hip abductors are strong and able to support body weight. (B) With L5 nerve root compression, hip abductors can be weakened, leading to positive Trendelenburg sign. This sign is seen when the iliac crest (pelvis) tilts down onto the side where the leg is raised, indicating contralateral hip abductor (L5) weakness.

risk factor for low back pain and a risk factor for a poor result after back surgery.^{35,36}

Physical Examination

Inspection

Inspection is the first step in the physical examination. As the patient walks into the examining room, gait should be observed. A sciatic list may be present, usually manifested as the patient leaning away from the side of leg pain. This sciatic list is thought to be associated with a paracentral herniation lateral to the nerve root. Axillary herniations may cause a list toward the side of herniation. The list is an attempt to relieve neuromeningeal tension by drawing the nerve root away from the herniated fragment. Another feature of gait that should be noted is a wide-based gait, indicative of lumbar or more cranial canal stenosis. A foot-drop or foot slapping gait may occur with L4 or L5 paresis. A Trendelenburg gait can suggest hip abductor weakness (Fig. 47.8), which may be a clue to L5 nerve root compression because the gluteus medius is most often an L5 dominant muscle.

Alignment of the spine should be noted. There can be loss of normal lumbar lordosis from muscular spasm. Hip flexion can relieve pain, leading some patients to lean forward or be reluctant to place the affected foot flat on the floor. In some cases, a nonstructural “sciatic scoliosis” can be noted on radiographs or examination of the back.

Palpation and Percussion

Examination should include a systematic examination of the back. The spinous processes are palpated individually and in stepwise fashion. Tenderness to palpation of one or two levels is more consistent with bony pathology than tenderness at multiple levels. In some cases, pressure on the spinous processes can reproduce sciatic symptoms. Continuing caudally, the lumbosacral junction and the sacral prominences and sacroiliac joint area can be palpated and percussed. Inflammation in these regions can manifest with sciatica, feigning disc herniation.

The musculature is examined next. The paraspinal muscles should be palpated. Spasm can be noted in addition to tenderness; this may be present as a “ball” of contracted muscle in one region. These findings are nonspecific and are of minimal diagnostic value. Muscular atrophy can be a sign of long-standing neural compression and is more likely isolated to one motor group. Generalized, symmetric muscle atrophy can suggest a more systemic neurologic disorder, such as a demyelinating disease.

Neurologic Examination

A neurologic examination is required in all patients with suspected herniated discs. Sensation of light touch is tested along dermatomes from L1 to S1. Standard dermatomal charts can be helpful, but there is variability among individuals; thus, this is highly subjective. In testing the upper lumbar roots, there is often a significant amount of overlap. The most discrete levels of testing are for L4, L5, and S1 nerve roots.³⁷ These nerve roots are the most often affected by lumbar disc herniations. L4 sensory function is tested at the medial ankle; L5, at the first webspace between the great and second toes; and S1, at the lateral aspect of the sole of the foot. Sensation is difficult to “grade.” It is more useful to document sensation as normal, diminished, or absent. Sensory function should be compared with the contralateral side because this may help detect differences. The examiner should be wary of the presence of a glove-and-stocking distribution sensory loss, which can indicate a peripheral neuropathy—such as that associated with diabetes, or functional overlay—as it is not anatomic.

The motor examination should proceed in a routine manner. In the lower extremity, it is better to test movements rather than specific muscles. S1 motor function is assessed by testing plantarflexion, whereas L5 is tested by toe dorsiflexion, particularly the great toe (extensor hallucis longus), and hip abduction. L4 involvement most often affects ankle dorsiflexion (anterior tibialis), although quadriceps function can be compromised. There is a significant amount of overlap of upper lumbar motor innervation. Knee extension can be considered L3 function (although L2 and L4 contribute); hip flexion, an assessment of L1–L2 function; and hip adduction, an assessment of L2 function. Motor function is graded as 0 to 5, with 5 being full strength against active resistance (Table 47.2). In particular, S1 function should be assessed by asking the patient to toe raise repeatedly or toe-walk. Because of the enormous strength of the gastrocnemius complex,

TABLE 47.2 Motor Strength Grading System by Physical Examination

Grade	Definition
0	No visible muscle contraction
1	Visible muscle contraction; no joint movement
2	Can move joint, but not overcome gravity
3	Able to overcome gravity, but cannot overcome any examiner resistance
4	Able to overcome some, but not full, examiner resistance
5	Full strength; able to resist full examiner force

even a weakened muscle can overcome the examiner’s hand. Toe-walking can show smaller differences, however, from side to side by using the weight of the patient’s body as the resistance. Repetitive toe raising may help detect smaller differences.

Deep tendon reflexes are tested at the patella and Achilles tendons. The patellar tendon reflex may be diminished or absent with L3 or L4 involvement, whereas the Achilles tendon reflex is affected primarily by S1. There is no specific reflex that reliably reflects L5 function. Reflexes are tested bilaterally and can be graded. Symmetrically decreased reflexes are not helpful in isolating a lesion. Generally, reflexes are anticipated to be decreased in response to nerve root compression from a herniated disc. Increased reflexes (hyperreflexia), especially if bilaterally symmetric, can indicate spinal cord compression at the thoracic or cervical level.

Specific Tests

The straight-leg raise (SLR) test is an extremely useful provocative test in examining patients with a herniated disc (Fig. 47.9). The classic test is performed with the patient in the supine position. The heel of a relaxed leg is cupped by the examiner’s hand and elevated slowly. The knee is kept in extension while the hip is flexed. The test is considered positive if sciatic pain is reproduced between 35 and 70 degrees of elevation. Studies have determined that in the first 35 degrees of elevation, the slack in the nerves is taken up, and at 35 degrees or more, tension is placed on the nerves. More than 70 degrees of elevation causes no further stretch of the nerve roots. The SLR test is best for eliciting L4, L5, or S1 radiculopathy. It is not useful for upper lumbar roots, for which a femoral stretch test should be used. A positive SLR test is indicative of nerve root compression in 90% of cases.³⁸ It does not implicate a herniated disc as the source of compression, however, because foraminal encroachment or other mass lesions can lead to a positive SLR test as well.

The SLR test should always be performed bilaterally. If raising the contralateral leg reproduces symptoms in the ipsilateral side, this is highly suggestive for a herniated disc. It should be considered strong evidence of the diagnosis and is more specific for a free disc fragment. The Lasègue maneuver is a modification of the SLR test. The leg is raised until radiating symptoms are produced. Then, the foot of the ipsilateral leg is maximally dorsiflexed. By increasing the tension along

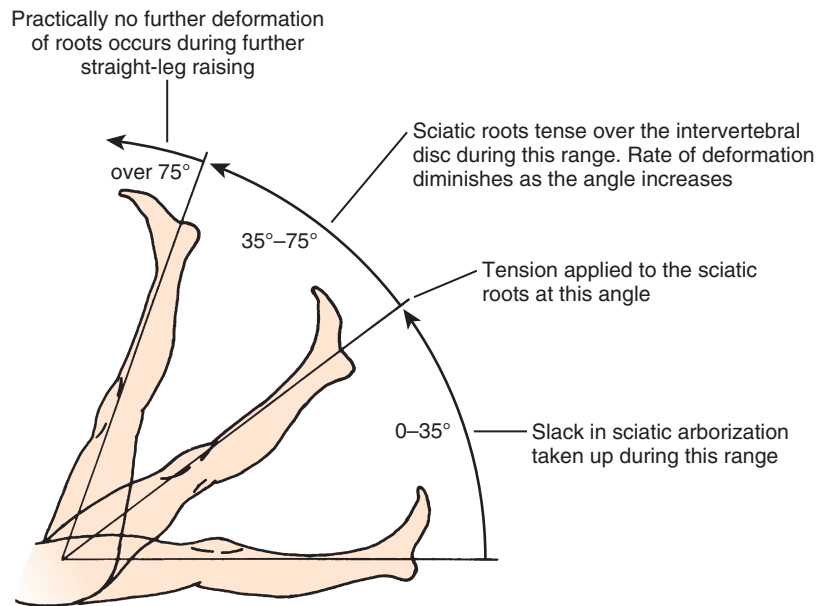


FIG. 47.9 Supine straight-leg raise test.

the sciatic nerve and lower lumbar nerve roots, dorsiflexion exacerbates pain and is considered a positive examination.

An important variant of the supine SLR test is the seated SLR test. When a patient is initially examined, the patient is usually seated at the side of the examining table with the knees and hips flexed at about 90 degrees. In this position, the heel is cupped, and the leg is extended at the knee. With a herniated fragment causing nerve root tension, the patient reflexively extends at the hip and leans back to relieve the ensuing sciatic pain. If the patient is comfortable with the seated SLR test but has a positive supine SLR test, symptom magnification must be considered because these findings are pathoanatomically contradictory.

The so-called slump test is a variant of the Lasègue test and SLR test. This test is performed in the seated position; the patient is asked to flex the thoracic and lumbar spine while fully flexing the neck. Next, the SLR test is performed while the foot is dorsiflexed on the same side, as denoted by the Lasègue test. The combination of these maneuvers adds cephalad gliding of the spinal cord to the examination, whereas the SLR test and Lasègue test by themselves produce only caudal tension on the nerve roots. A more recent study found that the slump test was more sensitive than the SLR test in patients with lumbar disc herniations, whereas the SLR test was more specific.³⁹

Another tension sign is the bowstring test. Starting with a typical SLR test, the leg is raised until symptoms are produced. The leg is flexed at the knee, and the tibial and peroneal nerves (distal aspect of the sciatic nerve) are placed on tension by palpation in the popliteal space. Reproduction of pain is considered a positive sign of root tension.

The femoral stretch test is performed in the prone position. The leg is flexed at the knee while pulling the hip into

extension. Reproduction of anterior thigh pain is indicative of upper lumbar root pathology.

Differential Diagnosis

The differential diagnosis should be narrowed based on history, physical examination, and selected imaging tests. Radicular pain can be caused by numerous compressive disorders, such as spinal stenosis, abscess, tumor, or vascular disease. Intrinsic nerve problems, such as nerve tumors or multiple sclerosis, can produce similar symptoms. Peripheral neuropathies, such as tarsal tunnel syndrome, meralgia paresthetica, and obturator or piriformis syndrome, can also lead to similar sciatic-type pain.

Diagnostic Imaging

We use a simple imaging algorithm for patients with suspected lumbar herniated discs. If the patient presents acutely, within the first 2 weeks of the incident, the examination is typically masked by a large amount of spasm, back pain, and generalized tenderness. If the mechanism of injury involved substantial trauma, plain radiographs are obtained. For low-energy injury, radiographs can be delayed until the follow-up examination at 6 weeks. Numerous patients will have recovered substantially by 6 weeks, obviating the need for further workup. For patients whose pain has not improved or perhaps has worsened, plain radiographs are obtained. Advanced imaging is reserved for patients in whom pain is persistent, the diagnosis is unclear, or surgical treatment is planned. “Red flags” in a patient’s history that should prompt early MRI include constitutional symptoms (i.e., fever, chills, and sweats),

a history of malignancy, osteoporosis, progressive neurologic deficits, or bowel and bladder incontinence.

Plain Radiographs

Plain radiographs cannot show a herniated disc. They can show changes that are suggestive of a herniated disc, however. As stated previously, a scoliotic list can be present on radiographs. This list may be convex or concave to the ipsilateral side and is not specific for a level.

Other findings that can be noted on plain films are changes consistent with disc degeneration, including osteophytes; disc space narrowing; or subtle changes in translation, facet hypertrophy, or changes in sagittal alignment. Most commonly, plain films are negative, especially in younger patients with an acute herniation. Plain films are important in ruling out obvious underlying problems, such as lytic lesions, tumors, infections, inflammatory spinal disorders, or instabilities (e.g., spondylolisthesis).

High-quality anteroposterior (AP) and lateral radiographs are prerequisites to planning operative interventions such as discectomy. It is crucial to recognize if there is an anomalous number of lumbar vertebrae, such as spines with a “lumbalized” first sacral segment (i.e., six lumbar vertebrae), because this can influence intraoperative identification of the correct disc level. Plain films can help detect other congenital anomalies, such as spina bifida occulta defects, which can influence surgical exposure and dissection.

Magnetic Resonance Imaging

MRI is the most popular modality for advanced imaging of lumbar disc herniations. MRI is superior to CT in delineating soft tissues. The disc and fragments that may have herniated from it are readily visualized. Free fragments (sequestered) can be differentiated from extruded disc herniations (Fig. 47.10), and a symmetric bulge can be differentiated from a contained protrusion. The neural elements themselves are well visualized. Neural encroachment can be detected within the spinal canal, the foramina, or extraforaminally. MRI is also useful in differentiating disc herniations from tumors, vascular anomalies, or bony compression.

Numerous features of a herniated disc can be noted on MRI. The size and type of disc herniation can be reliably determined using MRI, which may have prognostic significance.^{11,40,41} Carragee and Kim¹¹ correlated outcomes with herniated fragment size and its effect on canal area. Larger discs (>6 mm) were more likely to have a positive SLR test or femoral stretch test (Wasserman sign). In the operative group, larger discs were predictive of a better outcome. The fair and poor outcomes in operative patients were in patients with small discs (<6 mm).

Attempts to correlate MRI findings with clinical symptoms have been made. In 33 patients in whom disc herniation was diagnosed clinically and 5 control patients with low back pain alone, Kikkawa and colleagues⁴² performed three-dimensional MRI using a fast low-angle shot with gadolinium enhancement. Dorsal root ganglion enhancement was found to be



FIG. 47.10 T2-weighted magnetic resonance image has myelogram-like appearance, in which cerebrospinal fluid within dural sac is bright. This makes compression from disc herniations readily visible, as can be visualized in this case of a large extruded paracentral disc herniation.

nonspecific, occurring in controls and sciatica patients. Enhancement of the root proper was detected, however, in 11 of 30 symptomatic patients, with patients having a statistical tendency for more severe motor involvement. There was no significant association of diffuse versus local enhancement with the positivity of the SLR test or sensory changes. Central compression of the cauda equina did not lead to enhancement in any cases. Although these results are modest, they suggest a future use of gadolinium-enhanced MRI as a noninvasive method of determining the microvascular response to compression of neural structures.

Komori and colleagues⁴¹ studied the significance of enhancement around the herniated fragment itself. Patients with radiculopathy underwent initial and follow-up gadolinium-enhanced MRI to correlate clinical improvement with the degree of enhancement. Patients with marked decrease in size of the herniation showed good clinical resolution. This resolution was most significant in the “migrating”-type discs, which were closest to sequestered discs according to the authors’ description. Decrease in fragment size was associated with a gradual increase in the area of enhancement in 17 of 22 sequestered disc herniations, all of which had improvement of radicular pain. Five cases of sequestered discs without enhancement or size decrease had a poor clinical result. Enhancement was less marked in extruded versus sequestered herniations; however, herniations that did show enhancement had a significantly better clinical course. From these data, Komori and colleagues⁴¹ recommended this test as a prognostic tool in guiding the treatment of patients with extruded or sequestered herniated discs.

Of more recent interest is the influence of posture on the MRI appearance of discs and their relationship to the neural structures. Because images are traditionally acquired in the supine position, the spine is not axially loaded as it is during everyday activities. Weishaupt and colleagues⁴³ performed positional MRI in patients with low back or leg pain for 6 weeks that was not responding to conservative treatment. Images were obtained in the usual supine position and with a seated flexed and extended posture. Changes in foraminal size and neural compression occurred with flexion and extension. Changes in foraminal size correlated with increased pain scores. These findings are probably most significant for low-grade herniations (i.e., bulges or protrusions) in which there is still a fixed-volume system within the disc space provided by an intact outer annular layer. Similar findings have been shown using dynamic functional plain myelography.⁴⁴

Magnetic Resonance Imaging in the Postoperative Spine

Not all patients respond well to surgical discectomy. Failures may be related to numerous factors, including psychosocial disturbance, recurrence, infection, wrong-level surgery, poor surgical indications, and improper operative technique. Residual or recurrent back and leg complaints often prompt practitioners to seek postoperative imaging. Plain radiographs can show overall alignment, with flexion-extension views useful in detecting instability, spondylolisthesis, or disc space collapse, which is a frequent occurrence after discectomy in adult patients. Contrast-enhanced CT is best to show if there is associated bone or soft tissue impingement of the space available for the neural elements.

As the modality of choice for imaging the neural structures, MRI is frequently obtained. Because of edema, hematoma, and formation of surgical scar, MRI is best delayed until 6 months after surgery,⁴⁵ if symptoms allow. The main challenge is differentiating scar from new-onset disc. On standard T1-weighted sequences, this differentiation can be difficult. In the early days of MRI, T2-weighted images were not as useful because of longer scan times with inadequate magnet strength.⁴⁶

The superiority of MRI over CT to distinguish scar from disc has evolved with the introduction of gadolinium contrast agent; this is based on T1-weighted sequences and has been shown by many researchers. Opinions have been changing regarding this test as the gold standard. A herniated disc fragment may eventually enhance if enough time passes before it is imaged. This enhancement can lead to recurrent or residual disc fragments being interpreted as extradural fibrosis or scar. Enhancement of scar should occur within 15 minutes of injection.

Evidence suggests that sophisticated T2 image analysis might supplant the need for gadolinium-enhanced MRI. Barrera and colleagues⁴⁷ compared different imaging sequences with and without gadolinium contrast agent. These investigators documented 100% sensitivity for detecting scar for T2-weighted turbo-spin echo (TSE) and fluid-attenuated inversion recovery (FLAIR) sequences compared with T1-weighted images with gadolinium. Specificity was 94% and 92% for TSE T2 and FLAIR images. Barrera and colleagues⁴⁷ concluded that standard TSE T2 images acquired using a rapid

sequence are extremely sensitive and specific in distinguishing disc from scar in most cases, and that the use of gadolinium contrast agent should be reserved for the rare situation in which that distinction cannot be made. These recommendations are supported by others.⁴⁶

Grane and Lindqvist⁴⁵ studied the role of gadolinium enhancement of the nerve roots after discectomy. These investigators found intradural (within the cauda equina) nerve root enhancement in 59% of patients with recurrent clinical symptoms. Recurrent symptoms occurred, however, in 84% of patients with focal (extradural, after the nerve root has exited the cauda equina) enhancement and 86% of patients with nerve root thickening. Enhancement occurred in patients with and without evidence of nerve root displacement by scar or disc. This finding indicates that although symptoms may correlate with MRI enhancement, it is not associated with a compressive mass lesion.

In an early report on the use of MRI without gadolinium, Bundschuh and colleagues⁴⁸ studied 20 patients after failed disc surgery who had a strong likelihood of undergoing further surgery. In 14 patients, CT with contrast agent was also performed. The authors found that free fragments of disc had a mildly increased signal on T1 images compared with scar, whereas scar and disc were similarly hyperintense on T2 images. Overall, Bundschuh and colleagues⁴⁸ believed that MRI was at least comparable to CT with contrast agent in differentiating scar from disc, confirmed by intraoperative findings.

In general, patients requesting repeat imaging after a first episode of acute sciatica should be informed about the difficulty of MRI interpretation with respect to symptomatology. In a recent randomized clinical trial by Barzouhi et al., which compared surgery and prolonged nonoperative care for sciatica from lumbar disc herniation did not distinguish between patients with a favorable outcome and those with an unfavorable outcome.⁴⁹ Importantly, patients enrolled had an acute diagnosis of sciatica with a dermatomal pattern of pain distribution corresponding to the identified nerve root on MRI, and the character of the herniations were classified. Repeat MRI images at 1 year showed herniated discs in 35% of patients with favorable outcomes and in 33% of patients with unfavorable outcomes.

Myelography

Plain myelography previously was the imaging modality of choice in detecting herniated discs. It involves injection of intrathecal contrast material to outline the boundaries of the subarachnoid space and silhouette the enclosed neural elements. It is invasive and cannot show compression beyond the confines of the subarachnoid space. Extradural compression caused by a foraminal or extraforaminal disc can be missed. Advantages of myelography are that it is a dynamic test because images can be made with the patient standing.⁴⁴ Myelography should be reserved for cases in which noninvasive imaging, such as CT or MRI, are nondiagnostic, equivocal, or contraindicated. Currently, myelography is rarely used for the routine workup of herniated discs. When used, it is usually followed by a CT scan.

Computed Tomography

Before the advent of MRI, CT was the imaging modality of choice for evaluation of herniated discs. Using bone and soft tissue imaging techniques, herniations can be detected in various regions independent of the dural cavity. It has been shown to be 93% accurate in predicting surgical findings at discectomy. CT can also be performed with intrathecal contrast injection (CT myelography). We currently use this test for patients in whom MRI is contraindicated or cannot be obtained.

Some disc herniations can contain gas (Knutson phenomenon), noted on CT images. Mortensen and colleagues⁵⁰ reported four such cases that responded well to surgical discectomy. It is unknown if the gas forms before or after herniation. The clinical significance of the gas is not well understood. Ford and colleagues⁵¹ determined that intradiscal gas is composed predominantly of nitrogen.

Natural History

The key to enhancing one's skills as an intelligent diagnostician and patient advisor is understanding the available knowledge related to the natural history of degenerative lumbar disc disease. The exact natural history of lumbar disc herniations is variable and incompletely understood. A few well-performed natural history investigations are available. Some studies are of patients treated by various nonoperative methods. Others represent the nonoperative arm of operative versus nonoperative comparison studies.

In a widely quoted retrospective study, Saal and Saal⁵ found a 90% good or excellent outcome in patients treated nonoperatively for a lumbar disc herniation diagnosed by clinical examination and CT. Inclusion criteria were strict, including patients with SLR test positive at 60 degrees or less, leg pain greater than back pain, and electromyographic evidence of radiculopathy. Of patients, 92% returned to work. Nonoperative treatment consisted of aggressive physical therapy and back school education. A possible confounding factor is that many patients were referred for a second opinion regarding surgical versus nonsurgical treatment because they were anxious to avoid surgery. This may have introduced preselection bias error because the authors of the study were not surgeons. Concern has been raised about eventual fibrosis formation with nonoperative treatment of herniated discs. In a follow-up MRI study,⁶ the same investigators documented no increased risk for perineural fibrosis or adhesions with nonsurgical management.

Other authors have reported more modest results. In the nonoperative arm of Weber's¹⁰ classic randomized study, the long-term outcome of lumbar disc herniations was observed in 49 patients. Inclusion criteria were clinical signs and symptoms of L5 or S1 radiculopathy in addition to myelographic evidence of nerve root compression. Treatment included full-time bed rest for 1 week followed by partial bed rest the second week and back school instruction as an inpatient. At 1 year, 33% had good results, 49% had a fair result, and 18% had

a poor result. At 4 years, good results were reported in 51%, fair results were reported in 39%, and poor or bad results were reported in 10%. Because the tiered system is slightly different than that used by Saal and Saal,⁵ a direct comparison of the studies is difficult. If Weber's good and fair results are equated to Saal's excellent and good results, an 89% success rate achieved in the former at 4 years may be considered comparable to the latter's 90% success. Many of Saal and Saal's patients ultimately dropped out of the study and underwent surgical discectomy; thus, their 90% success rate might represent an overestimation.

In another nonoperative arm of a comparative study, 10-year follow-up results from the prospective Maine Lumbar Spine Study showed 61% improvement in the predominant symptom, 40% resolution of low back symptoms, and 56% satisfaction rate.⁷ Work and disability status were comparable between operative and nonoperative groups in this investigation. Similar findings were reported for the observational cohort of the Spine Patient Outcomes Research Trial (SPORT).⁸

In a subgroup analysis of SPORT data, obese patients, defined as having a body mass index (BMI) greater than or equal to 30 kg/m², were found to have less clinical benefit from both operative and nonoperative treatment. More important, the benefit of surgery over nonoperative treatment was not affected by BMI.⁵² McGuire et al. found a greater surgical treatment effect for highly obese patients primarily due to the poor outcomes of nonoperative management.⁵³ Rihn et al., in an analysis of the as-treated SPORT data, found that duration of symptoms affected outcomes following the treatment of lumbar disc herniation.⁵⁴ For patients with symptoms present for more than 6 months prior to seeking treatment, outcomes of nonoperative treatment were statistically significantly worse than those with symptom duration less than 6 months.

Methods of Nonoperative Treatment

Physiotherapy

Bed rest should be limited to no more than 2 to 3 days.⁵⁵ Greater periods of inactivity can potentiate prolonged disability and continued or augmented pain. Exercise therapy and physical rehabilitation should be included in the nonoperative care of herniated discs. Treatment goals are to restore strength, flexibility, and function that were lost secondary to pain, splinting, and spasm. Postural education to avoid activities that can increase intradiscal pressure or neuromeningeal tension, or both, should be provided.

Various regimens have been advocated, with none clearly superior. In our practice, physical therapy prescription usually includes torso stabilization training; paraspinal muscle stretching and strengthening; and a focus on gluteal, hamstrings, and abdominal exercises. These muscles are important in the static and dynamic stabilization of the spinal column. Some authors suggest concentration on flexion or extension maneuvers. In our experience, it is difficult, however, to predict if one or both of these movements would aggravate pain or help. Provisions for either flexion or extension concentration, at times combined with lateral shifts, are best left

to the therapist's assessment because the therapist can actively assess what is provoking pain. Concomitant facet arthritis, painful disc degeneration, and muscular pain can influence pain patterns and aggravating movements.

Adjunctive modalities can aid in relieving some associated symptoms. These modalities include ultrasound treatment, electrical stimulation, and massage. These may be helpful in short-term, symptomatic relief of back pain. Traction is also commonly prescribed. It theoretically may diminish intradiscal pressure, increase foraminal dimensions, and possibly relieve radicular pain secondary to herniated discs.⁵⁶ The role of chiropractic manipulation is controversial. Although some patients believe that manipulations have "reversed" the herniation, there is no evidence to support the ability of chiropractic manipulation to alter the normal or pathologic morphology of the disc.⁵⁷

Pharmacologic Treatment

Medications can be useful in decreasing disc-related symptoms. Because a local or systemic inflammatory reaction may participate in pain generation, antiinflammatory agents are believed to be beneficial. Nonsteroidal antiinflammatory drugs (NSAIDs) are first-line agents. Numerous choices are currently available, including cyclooxygenase-2 inhibitors. In prescribing NSAIDs, several important issues must be considered. The medications can have side effects. A history of gastrointestinal bleeding or peptic ulcers can be a contraindication to NSAID use. Although this risk is reduced with cyclooxygenase-2 agents, it is not nil, and the patient should be warned of the possibility. Additionally with NSAIDs, other systems, such as the kidneys and liver, may be detrimentally affected.

In the acute setting, back and radicular pain can be severe. Short-term narcotic use, such as a single dose of a morphine-derivative analgesic, can be useful. Narcotics should not be prescribed or administered in an extended manner; they should preferably be limited to a 2- to 3-day course. Also in the acute setting, a tapering dosage regimen of oral steroids can be helpful in decreasing inflammation-generated pain from nerve root irritation. In a randomized, double-blind, placebo-controlled trial of patients with acute sciatica due to herniated disc, Goldberg et al. demonstrated that a 15-day tapered dose of oral prednisone, compared with placebo, resulted in statistically significant, though modest, functional improvement without improvement in pain.⁵⁸

So-called muscle relaxants are frequently prescribed. Although the drug class name implies a direct muscular effect, these medications have more significant sedative effects. Medications such as diazepam and methocarbamol should be used sparingly. Truly antispasmodic medications such as baclofen or cyclobenzaprine can have a more direct effect on muscle spasms.

Selective transforaminal steroid injections can produce symptomatic relief in many patients. In our treatment protocol, injections are offered to patients who have failed noninvasive measures but either are not interested in or are not good candidates for discectomy. In patients with more than one

level of herniation, selective nerve root injection can be useful in determining the symptomatic level. That said, similar efficacy in pain relief between fluoroscopically guided epidural steroid injection and local anesthetic injection has been demonstrated in a recent double-blind, randomized controlled study.⁵⁹

In a prospective series touted as a natural history study, Bush and colleagues⁶⁰ reported the results of 159 patients with CT-confirmed disc herniations treated with epidural steroid injections. Although 91% avoided surgery, this underscores a glaring problem of so-called natural history studies in that an interventional treatment was used. Although its exact biochemical effects are still being elucidated, steroid injections may be effective in "avoiding" surgery. In an analysis of SPORT data, Radcliff et al. noted that a higher percentage of patients randomized to surgery who subsequently received an epidural steroid injection switched to the nonsurgical treatment group (41%) compared to patients randomized to have surgery who did not have an epidural steroid injection (12%).⁶¹ The authors concluded that this is confounded by the higher baseline desire to avoid surgery in the group of patients who received epidural steroid injections.

With TNF- α being considered a significant contributor to radicular symptoms associated with lumbar disc herniation,⁶² recent attention has been paid to delivering anti-TNF- α medications into the epidural space. Freeman et al conducted a multicenter, randomized, double-blind, placebo-controlled phase IIa clinical trial to examine the efficacy of etanercept, a TNF- α inhibitor, injected transforaminally to treat symptomatic disc herniation.⁶³ Compared to placebo, etanercept was found safe and effective in providing significant therapeutic relief for patients with sciatica. When compared to epidural dexamethasone injection, etanercept has shown similar clinical efficacy in back and leg pain reduction, with no significant difference between the two medications.⁶⁴ There has also been recent interest in the use of transforaminal epidural clonidine injections, for its potential to dual action as an antiinflammatory and analgesic,⁶⁵ though this remains controversial.

In a well-designed retrospective study, Wang and colleagues⁶⁶ studied 69 patients who failed nonoperative (or, more accurately, noninvasive) care and had requested surgery as treatment. Instead, each patient was advised to undergo one or more transforaminal steroid injections at the affected root level. Of patients, 77% had clinical resolution and had not undergone surgery at an average follow-up of 1.5 years. In agreement with other studies, clinical success was not related to disc size, percentage canal compromise, or degree of motor weakness. From these findings, selective nerve root steroid injections seem to produce at least short-term relief of radicular symptoms from a herniated disc.

Operative Versus Nonoperative Treatment

The results of operative and nonoperative treatment for symptomatic herniated discs have been compared in numerous studies. The Maine Lumbar Spine Study group published

1-year and 5-year results of an ongoing comparison of surgically and nonsurgically treated patients.^{1,67} More than 500 patients were included in the prospective observational study without stringent clinical or radiographic criteria except for disc-related sciatica treated with at least 2 weeks of nonoperative care within 2 months of onset. The decision to undergo surgery was determined on an individual basis and was not randomized. At 1-year follow-up, surgically treated patients were less symptomatic than patients in the nonoperative group, despite the former being *more* symptomatic at initial presentation. Relief of back or leg pain was reported by 71% of operated patients compared with 43% of the nonoperative group. High satisfaction levels and improved quality of life were documented for the operative group. For the workers' compensation group, there was no difference in time to return to work between operative and nonoperative groups. A criticism of the study is a substantial attrition rate, with 24% of patients unavailable for final follow-up.

In the 5-year outcome report, 70% of surgical patients reported back or leg pain improvement, whereas 56% of nonoperatively treated patients reported improvement.¹ As with the 1-year results, a similar percentage of patients were receiving workers' compensation benefits in both groups with no difference in return to work at final follow-up. Reoperations were performed in 20% of the operative patients; 16% of patients initially treated nonsurgically went on to operation. The authors noted that the benefits of surgery versus nonoperative treatment were greatest in the early part of the study, within the first 2 years, and that at final follow-up these advantages were less apparent.

Most recently, the Maine Lumbar Spine Study Group published their 10-year follow-up results.⁷ Of the eligible patients initially enrolled, data were available for 85% of patients treated surgically and 82% of patients treated nonoperatively. A significantly larger percentage of surgical patients reported relief of low back and leg pain than patients treated nonoperatively. Similarly, surgical patients exhibited better function and satisfaction compared with nonoperative patients. Nonetheless, improvement in dominant symptoms was reported for both treatment groups. Work and disability status were similar for both groups.

The 2006 SPORT investigation was designed to be a rigorous, randomized, prospective, controlled study.^{8,9,68} In its execution, there was a high rate of crossover between operative and nonoperative groups and a substantial portion of patients not willing to be randomized, resulting in a large observational group. The randomized and observational arms of the investigation reported improvement in bodily pain, physical function, and Oswestry Disability Index (ODI) scores regardless of intervention. In the randomized arm of the study, between-group differences favored surgical intervention, but these differences did not reach statistical significance in an intention-to-treat analysis. With the difficulties encountered with non-adherence to assigned treatments (i.e., crossover), the intention-to-treat analysis may not be reflective of the true outcomes. In an as-treated analysis, surgical treatment showed statistically superior results compared with nonoperative treatment. Similar results were found in the observational arm

of SPORT. These results have proven to be durable, as there has been little to no degradation of outcomes in either the nonoperative group or the operative group in 8-year follow-up reports.⁶⁹ With the analysis adjusted for the treatment received, operated patients still maintained clinically significant greater improvement in all primary outcomes compared to the nonoperative group, as was reported at earlier follow-up time points.

The classic work by Weber¹⁰ reported similar results to these findings. As briefly discussed in the prior section on natural history, Weber¹⁰ compared surgery versus nonoperative care in a randomized, prospective study. There were three study groups: one group "required surgery"; another showed no indications for surgery; and the third was the "undecided" group, in which it was unclear if surgery would be beneficial. Only the third group was randomly assigned to surgical or nonsurgical treatment.

At 1 year, good results were reported in 33% of the nonoperative group versus 66% in the surgical group; at 4 years, 51% versus 66%; and at 10 years, 55% versus 57%. These findings show that operative and nonoperative treatment outcomes seem to converge with time, being nearly the same at 10 years, and that the benefits of surgery are early. The 5- and 10-year results from the Maine study are similar, although the larger numbers of the Maine study achieved statistical significance. Both studies are flawed by patient selection bias. Weber randomly assigned only patients in the "unclear" group, whereas the Maine group had no specific criteria and included patients per individual physician investigators' usual practice.

Alaranta and colleagues⁷⁰ also prospectively compared operated versus nonoperated patients. In contrast to other similar studies, the nonoperative group was subdivided into cases with and without myelographic evidence of nerve root compression, whereas all operated patients had a positive myelogram. At 1-year follow-up, 91% of operated and 82% of nonoperatively treated patients with positive myelograms had improved pain levels. Only 51% of nonoperatively treated patients with negative myelograms had improvement. These data suggest that a distinctly worse natural history exists for patients with sciatica and no evidence of root compression. Although not specified by the investigators, this group probably represents sciatica from an extraspinal or non-disc-related origin. In a companion study published by the same authors, they further identified this group as having a high incidence of generalized pain (e.g., concomitant occipital headaches), more physically strenuous jobs, and lower pain thresholds.⁷¹ This study could possibly point to a preponderance of psychosocial and behavioral factors involved in the perpetuation of the symptom complex.

Operative Treatment

Indications

An absolute indication for lumbar discectomy is a progressive neurologic deficit. In this circumstance, operative intervention may be considered conservative care, provided that no medical contraindications exist. Progressive neurologic deficit

is most commonly associated with a cauda equina syndrome and is discussed in more detail later. The relative indications for discectomy vary among surgeons and patients. Discectomy, in its many shapes and forms, can produce symptomatic relief in appropriately selected patients. It is the surgeon's obligation to identify the patients in whom the anticipated benefits outweigh the attendant risks of surgery.

A prerequisite is radiologic identification of compressive pathology that is concordant with the patient's physical signs and symptoms. A patient with a large left L4–L5 level who presents with leg pain that radiates to the dorsum of the foot, weakness of toe dorsiflexion, decreased sensation in the first dorsal webspace, and a distinctly positive ipsilateral and contralateral SLR is an ideal candidate after failure of appropriate nonoperative treatment. In this “ideal” situation, the clinical and radiologic findings point toward compression of the L5 nerve root. To idealize the presentation further, the patient has a strong desire to return to work; is not involved in litigation, disability, or workers' compensation issues; and does not have any psychological issues. Important questions remain unanswered even in such cases, however, including optimal surgical timing, the method of disc excision, and the postoperative rehabilitation protocol.

Only a few patients match such textbook descriptions. Most patients lack one or more of the supportive diagnostic clues, making it more difficult to support the decision to operate. This situation does not represent a contraindication to surgery, however, because many published series show approximately 85% success rates in patient groups with lesser percentages of objective motor and sensory findings.^{3,38,67,72–79} The slim chance of back pain relief in relation to leg pain with discectomy that surgeons obligatorily confess to their patients is also probably an underestimation because most series document at least modest improvements in back pain. It is incumbent on clinicians to discuss the advantages, disadvantages, risks, alternatives, and estimated expected outcomes with patients.

Available Techniques

A vast array of techniques exist for surgical treatment of herniated discs.^{80–85} Standard open discectomy is the most common surgical approach.^{72,77,86,87} It involves careful incision planning, laminotomy or partial laminectomy to provide adequate visualization of the pathology, gentle retraction of the neural elements, and direct excision of the herniation. As an adjunct to open discectomy, some surgeons advocate the use of a microscope for better visualization and minimizing incision size.^{31,79,83,88,89} The purported advantage of the microscope is the ability for the surgeon and the assistant to visualize the operative field equally through a smaller surgical wound.

Alternatives to interlaminar techniques have been developed for excision of foraminal and extraforaminal lateral disc herniations, which involve exposures between the transverse processes and lateral to the pars interarticularis.^{14,89} Although some surgeons have advocated the addition of fusion,^{76,90} this practice is unpopular.⁸¹ Advocates hold that fusion decreases

the chance for reherniation; however, only a complete discectomy with interbody reconstruction can eliminate this risk. Long-term effects such as an adjacent-segment degeneration and the additional morbidity and complication rates are potential disadvantages.^{76,81,90,91}

With increased interest in smaller incisions and minimally invasive surgery, various percutaneous methods of treatment have been developed. Some methods entail placement of a cutting device intradisally to decompress the disc space to retract the herniated fragment.^{92–95} Other methods involve percutaneous techniques of directly visualizing the neural elements and disc using an endoscope.^{80,96} Chemical digestion of the disc (i.e., chemonucleolysis) had enjoyed popularity in the past. However, enzyme-related complications and results inferior to open discectomy limited continued popularity in the United States.^{92,97,98}

Repair of the annulus fibrosus following lumbar discectomy is an emerging technology, with various techniques and devices introduced in recent years. There remains a paucity of published studies concerning the clinical outcomes of newer devices, though data are forthcoming. In one published prospective randomized trial, an annular repair technique did not show statistically significant benefit in patient symptoms, but did reduce the need for subsequent surgery.⁹⁹

Open Simple Discectomy

Timing

Disagreement exists regarding the optimal timing for surgery. The question is how long should nonoperative care be continued (without improvement) before the outcomes of surgery are detrimentally affected? Stated another way, in patients who are not responding to nonoperative measures, when should surgery be performed? In evaluating the currently available surgical results, this interval may be 2 to 16 months. The clear exception to delaying surgery is a patient who is experiencing a progressive neurologic deficit or cauda equina syndrome.

Rotheorl and colleagues³³ stratified operatively treated patients according to time from presentation to surgery. Patients with symptom duration of more than 2 months had a statistically significantly worse outcome than patients operated on within 2 months. There was no difference if surgery was performed within 1 or 2 months. Likewise, Hurme and Alaranta¹² found the best results in patients operated on within 2 months of the onset of disabling sciatica. Nygaard and colleagues³² reported worse results in patients with leg pain for 8 months or more; Jansson and colleagues³⁶ found similar results in their analysis of the Swedish registry data, which included greater than 27,000 patients. Sorensen and colleagues³⁴ found that symptom duration greater than 16 months was predictive of poor results, but this was highly influenced by patient personality and social factors. The disparity between these findings is difficult to explain. Randomized prospective data comparing early versus late surgery are lacking. In our practice, surgery is performed within 6 months from symptom onset if nonoperative care has failed. The important message

is that there is not an urgency to perform surgery. Nerve roots, as compared with the spinal cord, are fairly resilient.

Other factors can influence the time to surgery. Ito and colleagues⁴⁰ found that patients with uncontained herniations had surgery much earlier than patients with contained herniations. Specifically, 56% of patients with uncontained herniations and 21% of patients with contained herniations had surgery within 1 month. This finding was not correlated to outcomes but was influenced more by the severity of symptoms. Early surgery does not seem to affect the rate of neurologic recovery; objective improvements in motor and sensory deficits do not seem to correlate with symptomatic relief and overall success rates.¹⁰⁰

Technique

General endotracheal anesthesia is induced. Alternatively, spinal anesthesia can be used. The patient is carefully logrolled into the prone position. An Andrews frame can be attached to a regular operating table to facilitate the prone kneeling position. Special spine frames, such as the sling attachment for the Jackson (OSI) table (Mizuho OSI, Union City, CA), can be used. These frames allow hip flexion to produce some flexion of the lumbar spine, which widens the interlaminar space (Fig. 47.11). The kneeling position also allows the abdomen to hang free, which helps indirectly to reduce the epidural venous pressure and reduce intraoperative blood loss. One set of hip pads is placed just distal to the anterior superior iliac crest. Adequate padding protects against lateral femoral cutaneous nerve injury. The thighs are allowed to flex at the hip, with the knees resting within the sling. The sling should be positioned low enough to allow adequate hip and knee flexion and is padded with gel pads and pillows. A pad is placed between the knees and medial malleoli to prevent pressure necrosis. The transverse chest pad is placed just above the xiphoid process.

With either table, the arms are placed in the 90-90 position with the cubital tunnels protected. The advantages of the Jackson table include that it allows a one-step turn to the prone position (avoiding the second step of pulling the patient's buttocks to the posterior pads with the Andrews frame) and its complete radiolucency. The disadvantage is that the table cannot be rotated away from the surgeon to allow better visualization by the assistant. With the patient secure, the superior aspects of the crests are palpated and marked; this usually corresponds to the level of the L4-L5 disc space or the L3-L4 interspinous process interval. The midline spinous

processes are marked, and the interspinous regions are counted from the sacrum up. The target interspinous space can be indicated with a marker or scratched in the skin using a sterile 25-gauge needle after alcohol skin preparation. If the patient is unusually large and landmarks cannot be palpated, a radiograph can be taken to determine the level of skin incision before preparation. The lumbar region is then prepared and draped. A 3- to 5-cm midline skin incision is made, and can be extended if necessary in obese patients.

The subcutaneous tissue is dissected down to the level of the lumbar fascia. The spinous processes are palpated in the midline. The deep lumbar fascia is incised, on the side of the herniation, adjacent to the spinous process. The fascial incision spans the spinous processes of the adjacent segments. An intraoperative lateral radiograph should be obtained at this time to confirm the correct level. A marker, either a Kocher clamp or a spinal needle, can be placed along the interspinous ligaments. On the radiograph, the marker should point toward the operative disc space. It should angle superiorly in line with the interspinous process space.

The paraspinal muscles are subperiosteally elevated from the lateral aspect of the spinous processes using electrocautery. It is usually necessary to expose only the superior and inferior aspects of the adjacent segments. At the junction of the spinous process and lamina, blunt dissection is performed laterally along the interlaminar space using a large Cobb elevator. Care is taken to expose, but not violate, the facet joint.

A Taylor retractor is positioned to maintain exposure of the interlaminar space. The tip of the retractor is inserted lateral to the facet joint, and the instrument is levered laterally. A roller-gauze is looped underneath the surgeon's foot and tied to the handle of the Taylor retractor. Appropriate foot leverage maintains its position. Alternatively, numerous small retraction systems can be used. Meticulous hemostasis is achieved with a bipolar electrocautery. Extraneous posterior muscle and soft tissues that impair vision should be removed using a large curette and rongeur. The interspinous ligaments should not be disrupted.

A medium-sized curette is used to detach the ligamentum flavum from the inferior aspect of the superior lamina (Fig. 47.12). Because the ligamentum inserts along the anterior aspect of the lamina, the curette needs to be introduced at an angle with the spoon facing cephalad. The instrument is worked from side to side to release all layers of the ligamentum. In some cases, it is difficult to release the ligamentum flavum completely without removing a portion of the inferior

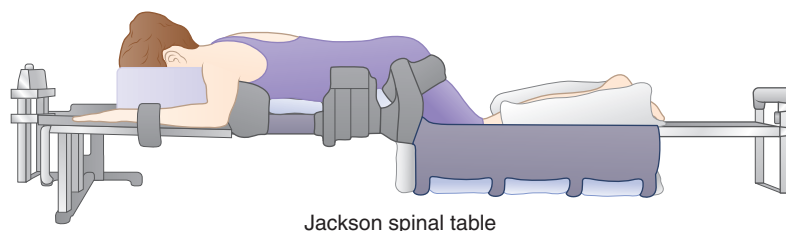


FIG. 47.11 The Jackson (OSI) table can be used to place the patient into a kneeling position, increasing interlaminar lumbar space while allowing the abdomen to hang free.

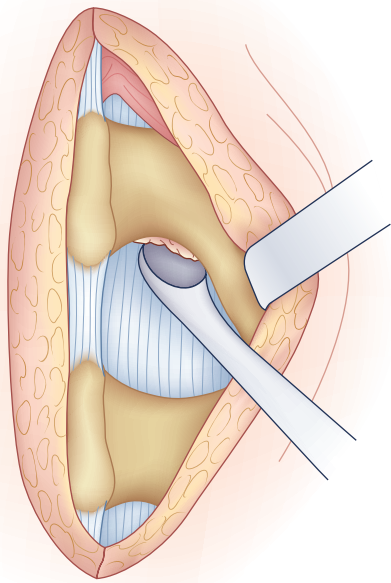


FIG. 47.12 The ligamentum is released from the inferior aspect of the superior lamina. The cup of the curette should face away from the dural sac to avoid inadvertent injury or tear.

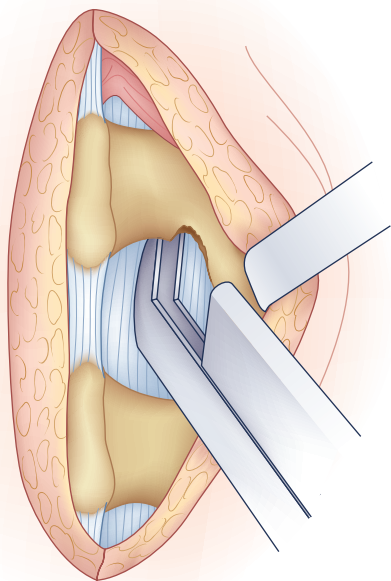


FIG. 47.13 In some cases, particularly above the L5–S1 level, a small portion of lamina must be removed to release the ligamentum flavum fully.

hemilamina (Fig. 47.13). Bone resection can be kept to a minimum but should not be avoided if it jeopardizes the exposure of the cauda equina and nerve root.

After the ligamentum is released superiorly, an angled probe, such as a Woodson elevator, can be inserted just deep to the ligamentum flavum (the yellow ligament). The Woodson

elevator is positioned along the periphery near the bone. A long-handle No. 15 blade scalpel is carefully used to incise the flavum directly on top of the Woodson elevator, which protects the dural sac deep to it, which may often be displaced posteriorly by a large herniated disc. The blade should not be inserted deep to the ligament or the elevator. If concerns exist, the ligamentum flavum should be incised layer by layer with consecutive passes of the blade until the Woodson elevator pops through; this is continued along the periphery of the interlaminar window. Optionally, the medial aspect of the ligamentum flavum can be left attached, and the ligamentum can be flapped open. At the completion of the discectomy, the flap can be replaced to act as a barrier against epidural adhesions.

With the ligamentum retracted or removed, epidural fat is visualized. In some cases, the fat layer has thinned from a displaced or inflamed nerve root. If present, the epidural fat is swept away from the dura with a Penfield No. 4 elevator. The focus of this maneuver is to identify the lateral aspect of the exiting nerve root and cauda equina. Epidural vessels are coagulated using irrigating bipolar cautery, as necessary.

A medial facetectomy often must be performed to visualize the nerve root adequately. A 3-mm or 4-mm Kerrison rongeur can be used in a back-hand manner to resect the medial aspect of the facet joint. Excessive resection should be avoided because the integrity of the facet and pars interarticularis is crucial to maintaining stability. It is wise to protect the underlying dura by pointing the Kerrison rongeur in the direction of the exiting nerve root. The endpoint of resection is adequate visualization of the shoulder of the descending nerve root adjacent to the medial border of the pedicle.

Before discectomy, a blunt-tipped probe is passed along the root and out the foramen to assess the amount of space available. Using a Penfield No. 4 elevator, the nerve root is mobilized medially. This mobilization can be facilitated using the bipolar cautery to lyse adhesions between the dura and the posterior disc. Retracting the nerve root medially shows the posterior anulus or the herniated fragment, or both. A nerve root retractor is used to maintain retraction (Fig. 47.14).

The disc fragment may be extirpated using the Penfield elevator to ensure that it is free of dural adhesions. With clear visualization of the fragment and its distinction from the neural elements, a pituitary rongeur can be used to explant it. If the fragment is extruded or sequestered, one can often identify a tear in the posterolateral or central anulus. If the herniation is subligamentous, a No. 15 blade scalpel can be used to create a small annulotomy to gain access to the herniated nuclear material. A pituitary rongeur can be carefully inserted into the annular defect to remove the fragmented disc.

Caution must be used when removing an axillary disc herniation. The fragment is easily visualized, tempting the surgeon to proceed directly with excision with the pituitary rongeur. Removal of large fragments that extend deep to the cauda equina and nerve root can lead to excessive bleeding from the transverse anastomoses of the epidural veins in the axillary region. It is preferable to retract the root medially and remove the fragment from a lateral approach, if possible.

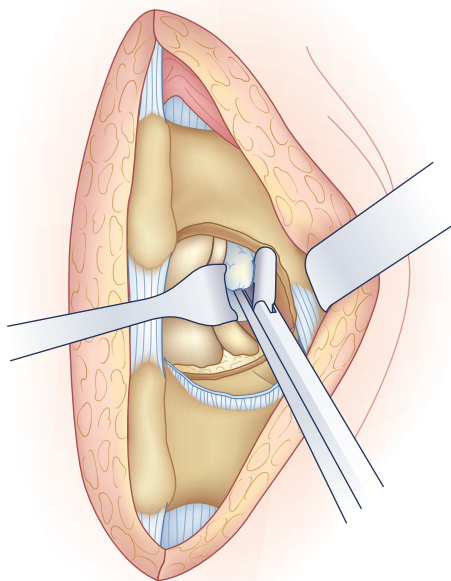


FIG. 47.14 A nerve root retractor is used to retract gently the descending nerve root and cauda equina toward the midline to visualize the disc space and herniation.

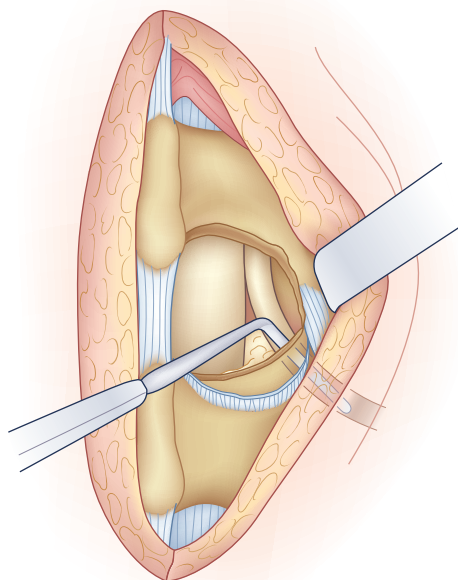


FIG. 47.15 After the herniated fragment has been removed, the spinal canal should be systematically inspected for any remaining fragments. A Woodson elevator is used to ensure that the neural foramen and nerve root are free. If the root is not completely free, additional fragments or compression may be present.

The mobility of the nerve root is assessed using a Penfield No. 4 or Woodson elevator (Fig. 47.15). After discectomy, the nerve root should be freely mobile and under no tension or compression. If it is not, additional pathology should be suspected. Using a Woodson elevator, the epidural space must be

systematically inspected in a clockwise fashion, checking the canal ventral to the thecal sac and the patency of the foramen. A concomitant lateral recess or foraminal stenosis from bony encroachment is decompressed with Kerrison rongeurs. Finally, the disc space is irrigated with normal saline to dislodge any remaining loose fragments.

If the ligamentum flavum was spared, it can be loosely replaced. A fat graft placed over the dura does not improve clinical results but does facilitate reexploration and theoretically prevents adhesions from inhibiting normal neuromeningeal motion.^{101,102} The wound is copiously irrigated. Adequate epidural hemostasis is ensured before closure. In most cases, a drain is not needed. All retractors are removed, and the deep lumbar fascia is approximated with interrupted No. 1 absorbable suture. The subcutaneous layer is closed in two layers, and the skin is approximated with a subcuticular stitch or staples. The wound is cleaned, and a sterile dressing is applied.

Postoperative Care and Rehabilitation

After an uncomplicated simple open discectomy, the patient is discharged on postoperative day 1 or 2. Outpatient discectomy can be performed on selected patients with appropriate presurgical planning and postdischarge discussion.⁸⁶ The level of activity in the first week after surgery usually is limited by incisional pain. If leg symptoms were predominant preoperatively, the patient typically reports immediate relief.

The activity level recommended after surgery varies among surgeons. Concerns are that aggressive movements and loads can predispose to reherniation or excessive scarring. This concern leads many surgeons to limit lifting and bending after discectomy for about 3 to 4 weeks. Although this is probably the predominant practice, there is little literature to support such extended periods of protected activity.

Unrestricted activity protocols have shown success rates comparable to other series. Carragee and colleagues⁸⁷ conducted an uncontrolled, prospective trial of 152 patients who were allowed full activity after lumbar discectomy. Approximately one-third of patients returned to work in 1 week, with most (97%) returning to work within 8 weeks. Recurrent sciatica occurred in 17 patients and was considered possible reherniation (11%). Nine of these patients experienced improvement, and eight underwent repeat surgery. This rate is similar to reherniation rates previously reported, which have ranged from 0% to 13%. Carragee and colleagues⁸⁷ concluded that an unrestricted activity protocol does not negatively affect results after discectomy.

Other studies have focused on possible advantages to an early rehabilitation protocol after discectomy. Kjellby-Wendt and Styf¹⁰³ performed a prospective, randomized, controlled study comparing an early active versus traditional training program. The focus of early activity was to reduce lumbar edema and maintain mobility of the neural elements through motion and trunk-strengthening exercises. A pain-coping program was included in the early active group that was not included in the traditional group. There was greater range of motion and decreased pain at 12 weeks in the early group.

At 1-year follow-up, both groups were equivalent. Patient satisfaction was ultimately greater in the early active group (88%) versus the control group (67%), although this was not statistically significant. In critique of the study, it was unclear why the difference in subjective outcome was not accompanied by sustained objective benefits. Inclusion of the pain-coping program in the study group, which would have minimal influence on range of motion, may have positively influenced patients' perception of their residual impairment. In support of the findings by Carragee and colleagues,⁸⁷ there was no difference in reherniation rates between the two groups.

Gencay-Can and colleagues performed a clinical trial comparing patients undergoing lumbar discectomy treated with a supervised aerobic exercise program starting 1 month after surgery in addition to home exercises with patients performing home exercises alone.¹⁰⁴ While there was no significant advantage in visual analog scale (VAS) back and leg pain, there was significantly better functional improvement in the aerobic exercise group as measured by the Roland Morris Disability Index. As one may expect, the functional recovery did not carry forward once the exercise program was discontinued.

Alaranta and colleagues¹⁰⁵ performed a randomized trial comparing an "immediate" versus "normal" rehabilitation program. Physical activities, such as sports and games, were encouraged in the immediate activity group; psychological and social counseling were also provided. Details of the normal rehabilitation protocol were not provided. At 1-year follow-up, there were no differences in subjective outcomes, postoperative impairments, or return-to-work statistics. These results further indicate that early activity programs do not seem to be harmful but offer only minimal, if any, long-term benefits. Our decision to initiate early return to unrestricted activity is made on an individual basis.

Outcomes

The outcomes of surgical discectomy are reliable when one adheres to strict preoperative selection criteria. Factors that can influence these outcomes, such as location of the herniation, preoperative psychological status, surgeon experience, and work status, have been extensively analyzed.

Carragee and Kim¹¹ correlated operative outcomes with herniated fragment size and its effect on the canal area. Using axial MRI, these authors recorded several parameters, including disc area, canal area, AP disc length, and AP canal length. Patients with larger discs (>6 mm) were more likely to have positive SLR or femoral stretch tests. Comparing patients with operative and nonoperative treatment, the former had larger AP disc lengths and larger ratios of disc to canal area. In the operative group, larger discs were predictive of a better outcome. In the nonoperative group, symptom duration less than 6 months before presentation, no litigation, and younger age were predictive of a better outcome. All of the fair and poor outcomes in operative patients were in patients with small discs (<6 mm). Summarizing these findings, it seems that larger discs respond to surgery better than small discs and

that disc size is less predictive of outcomes in the nonoperative group. A criticism of this study involves the existence of a selection bias because the decision to treat operatively was nonrandomized.

Knop-Jergas and colleagues¹⁰⁶ analyzed the results of lumbar discectomy based on anatomic location. They characterized herniations as central, paracentral, intraforaminal, extraforaminal, or multiregional broad-based protrusions. The best outcomes were documented with paracentral and intraforaminal discs, with 80% yielding good or excellent results. Central discs and multiregional discs had worse results, with 47% and 54% good or excellent outcomes. There was only one extraforaminal disc, which had a poor outcome. The level of herniation was not predictive of outcome, with 59% occurring at L4–L5.

In contrast, other authors have found that the level of herniation can affect operative outcomes. Dewing and colleagues¹⁰⁷ reported that discectomy for L5–S1 herniations had significantly greater improvements in VAS leg and ODI scores compared with L4–L5 herniations. These authors also found that patients with sequestered disc fragments had significantly better outcomes than patients with contained disc herniations. In a post-hoc analysis of SPORT trial data, Pearson and colleagues⁶⁸ examined the influence of disc herniation location (central or lateral) and morphology (protrusion, extrusion, or sequestration) on the results of operative and nonoperative treatment. Their data suggest that the advantages of surgical treatment over nonoperative treatment were greatest for patients with lateral disc herniations. The magnitude of improvement from surgical treatment was not related to the location or morphology of the herniation.

Various preoperative and postoperative findings may also be predictive of results. Barrios and colleagues⁷³ found that the use of (and positive response to) traction as part of preoperative conservative treatment was predictive of a good outcome with surgery. In agreement with other studies, patients with sedentary, non-physically demanding jobs had better results than patients with more strenuous occupations. Jansson and colleagues³⁶ found that reduced preoperative walking distance and a history of back pain greater than 6 months were predictors of lower functional outcomes after discectomy. Jonsson and Stromqvist¹⁰⁸ found a persistently positive SLR test to be a reliable indicator of inferior results; this was particularly evident in patients in whom the test was positive for more than 4 months after discectomy.

As part of an ongoing study of lumbar discectomy, the Maine group published an investigation comparing operative rates and outcomes between high versus low rate of surgery regions within the state.¹⁰⁹ The data suggest that, in the high-rate region, the results were inferior compared with the low-rate region. These calculations were based on the rate of surgery per capita (population). The investigators concluded that this difference was most significantly related to the surgical indications used by the individual physicians and that, in the higher-rate areas, the indications may have been less stringent. This might have been true, but an additional factor was not highlighted, which we calculated from the data. In the high-rate region, there were more surgeries done per capita,

but the number of operations done per physician was lower, averaging 11 surgeries per surgeon. In the low-rate region, surgeons averaged 26 operations each, although the overall rate of surgery was lower per capita. This factor directly supports previous analyses of the results of total joint arthroplasty, which indicate better results in high-volume hospitals because of concentrated experience. From these data, discectomy is optimally performed by an experienced, high-volume surgeon who employs strict indications.

Psychological and social factors have been shown by many investigators to influence profoundly the surgical results of lumbar discectomy.^{34,74} Sorensen and colleagues³⁴ found that preoperative psychological assessment was 86% predictive of surgical results after discectomy. Cashion and Lynch⁷⁴ found that patients with a good outcome were more self-confident individuals, were only mildly depressed, and were generally optimistic about the outcome of surgery. Slover and colleagues³⁵ found that a history of chronic headaches, smoking, depression, and self-rated poor health were associated with poor outcomes after discectomy. Jansson and colleagues³⁶ found that smoking was the most significant risk factor in patients who failed to improve after surgery. Olson et al. found that patients with higher education level demonstrated significantly greater improvement with nonoperative treatment, while surgical outcomes were not affected.¹¹⁰

With increased attention on the use of opioid analgesics in the United States, Radcliff et al. examined whether the use of these agents had an effect on outcomes for patients with lumbar disc herniation.¹¹¹ While patients who were treated with opioids had significantly worse baseline pain and quality of life, at 4-year follow-up, there was no long-term difference in outcome associated with opioid medication use. They also found that opioid use was not associated with surgical avoidance.

Spengler and colleagues¹³ calculated preoperative assessment scores for 84 patients before discectomy and correlated these with outcomes. The four components of the score were neurologic signs, clinical tension signs, psychological factors, and imaging evidence of neural compression. Imaging studies were most predictive of operative findings but not of outcome. The best predictor of outcome was the psychological score. This was based on the Minnesota Multiphasic Personality Inventory.

Hurme and Alaranta¹² prospectively studied 220 patients after discectomy, analyzing preoperative and perioperative factors. Optimistic patients had better results. Patients who preoperatively decided not to return to work had poorer outcomes. Patients who perceived their jobs to be physically strenuous had inferior results. The most predictive factors of a poor outcome were the decision not to return to work, marital status (divorced or widowed), age older than 40 years, a protracted period of sciatic pain, and multiple nonspecific somatic complaints. Predictors of good results were a high preoperative pain index, higher education level, overall satisfaction with life, and the perception that the patient's job was of light or suitable duty. Patients with a highly positive SLR test and younger patients with large disc herniations tended to have better results. This study highlights the importance of

psychological, social, and objective physical factors in predicting the outcome of lumbar disc excision.

Dvorak and colleagues¹¹² retrospectively applied a set of "accepted" operative indications—including radicular pain, positive SLR, contralateral SLR, dermatomal hypesthesia, motor deficit, and diminished deep tendon reflexes—to a series of patients who underwent discectomy. Of the patients, 65% fulfilled these criteria, and 35% did not. The long-term outcomes were not significantly different between these two groups. Patients who returned to work postoperatively were on disability compensation less than 2 months before surgery, whereas patients who did *not* return to work were on disability compensation an average of 4 months before surgery. Indications based on objective physical findings alone do not ensure a satisfactory outcome. The preoperative period of disability is a significant factor.

Some proponents of microscopically assisted discectomy purport that results are superior compared with standard techniques. In a published review of the literature by one of the strongest advocates of microdiscectomy, McCulloch⁸⁸ concluded that the available data are insufficient to show its superiority. He noted that successful results have ranged from 80% to 96%, regardless of the technique used, and highlighted the importance of patient selection as the more important determinant of outcome. Series of microsurgical discectomy report comparable rates of complications, such as dural tear and recurrent herniations.^{113,114} In a report with 5-year follow-up, long-term outcomes for minimally invasive techniques are comparable to those reported using standard techniques.¹¹⁵ It is important to note that radiation exposure to the surgical team and the patient can be 10 to 20 times greater with the use of minimally invasive techniques.¹¹⁶

There is disagreement regarding the efficacy of simple fragment excision (the so-called Williams sequestrectomy¹¹⁷) versus more extensive nucleus curettage. In a retrospective study of 200 patients, Faulhauer and Manicke⁷⁷ showed a lower reherniation rate with fragment excision alone. This comparison was inherently flawed. The standard discectomy group, by definition, had herniations that were in continuity with the disc space and could be characterized as subligamentous (contained). The fragment excision group obligatorily had displaced fragments that were not contained within the disc space. The most useful information perhaps is that when an extruded or sequestered (uncontained) fragment is found, acceptable results with a low herniation rate can be expected with fragment excision alone. Although aggressive methods of removal of further nuclear material may not be warranted, meticulous examination of all quadrants of the epidural space surrounding the nerve root and cauda equina should be performed in each case to avoid missing disc fragments.

Balderston and colleagues⁷² compared 40 patients who underwent simple fragment excision in one center with 40 patients who underwent excision and curettage in another center. The reherniation rate was not significantly different: 12.5% in the former group and 11.6% in the latter group. There was also no difference in the rate of disc space narrowing between the two groups. The only difference was a higher rate

of postoperative back pain in the curettage group at a minimum 2-year follow-up. In distinction to Faulhauer and Manicke's work,⁷⁷ the types of disc herniation were not different between the two groups.

Carragee and colleagues¹¹⁸ reported the results of a prospective, controlled study comparing subtotal versus limited discectomy. In the latter group, only the extruded disc material and loose fragments present in the disc space were removed. In the former, a more aggressive disc space curettage and debridement was performed in addition to removal of the extruded fragments. Early findings suggested a shorter convalescence in the limited discectomy group but decreased VAS and ODI scores in the subtotal group. These differences were not statistically significant at 2-year follow-up. Although the reherniation rate in the limited group was 18% compared with 9% in the subtotal group, this difference was not significant with the numbers available. Type II (beta) error cannot be excluded, however, because a significant difference may have been detected with greater patient numbers. The clinical significance of a 9% difference in reherniation rate may outweigh the lack of statistical significance.

Complications

Numerous complications can occur with lumbar discectomy, albeit at acceptably low rates. Recurrence rates range from 0% to 18%. In addition to differences in surgical technique and other factors, this broad range may reflect differences in the definition of recurrence. In some series, *recurrence* is defined as recurrent sciatica, whereas in others use of the term is limited to patients who required reoperation. The strictest definition is a true reherniation at the level and side previously operated, which ultimately leads to a frequency of 2% to 5%.

Wound infections have been reported in 0% to 3% of cases.^{12,72,77,84,86,117} These may be superficial or deep. Superficial infections may be managed with local wound care and antibiotics. Deep infections should be surgically debrided and irrigated. Epidural abscess is rare, with reported rates of 0.3%, and should be managed with surgical evacuation.¹¹⁹ The microscope has been considered a possible source of contamination because of the exposed unsterile optics that are in close proximity over the wound.^{88,120} Infection rates are comparable, however, to cases performed without the microscope.

Pyogenic discitis may occur after discectomy 2.3% of the time.^{12,119} Early detection is crucial in avoiding extensive bone involvement. Intravenous antibiotic therapy is usually successful, with rare cases requiring surgical debridement. MRI findings, including increased bone edema near the endplates and loss of disc space height, are difficult to discern from typical degenerative Modic-type changes. Laboratory evidence, such as elevated erythrocyte sedimentation rates and C-reactive protein levels, is important in confirming the diagnosis.

Vascular injuries are exceedingly rare. Injury to anterior vessels from perforation of the disc space have been docu-

mented in a few case reports.^{121,122} Arteriovenous fistula has also been documented.^{123,124} The most important component of managing these complications is prompt recognition and aggressive treatment, including vascular repair. Excessive epidural bleeding is uncommon, although one series documented blood loss of more than 300 mL in 4% of cases.¹²

Incidental durotomy occurs 0% to 4% of the time.^{8,72,84,113,125} It has been associated with a poor outcome in some series. At an average of 10 years after surgery, Saxler and colleagues¹²⁵ reported a lower rate of symptom resolution and a greater rate of chronic pain and headaches in patients who sustained an incidental durotomy. This study did not discern between primary and revision cases, which may have been a potential confounder. The potential for long-term clinical sequelae after incidental durotomy during discectomy is most likely influenced by the size of the tear, the ability to repair it, and the coexistence of neurologic injury.

Instability is quite rare after discectomy. Preservation of the facet joints is helpful in avoiding this sequela, although in our experience, a large percentage of the facet joint can be resected unilaterally without adverse effect. Some authors have indicated that instability can occur 30% of the time. This percentage is highly dependent on the definition of instability. Padua and colleagues¹²⁶ found radiographic evidence of instability, detected by flexion-extension films, in 20% of patients who underwent discectomy. Only 6% seemed to be symptomatic, however. In contrast, Faulhauer and Manicke⁷⁷ defined instability by clinical findings, such as apprehension with flexion or extension, instead of radiographic measurements. These authors reported rates of 16% and 30% with two different operative techniques; 3% had severe enough symptoms to warrant a brace, and only one patient eventually went on to fusion. Kotilainen and Valtonen¹¹⁹ found that the presence of clinical apprehension postoperatively was predictive of a poor result after surgery. These data highlight the importance of documenting preoperative radiographic and clinical instability so that the effects of the discectomy itself can be better assessed in the postoperative period.¹²⁷ Biomechanically, simple discectomy is not a destabilizing procedure if performed properly.

Recurrent Disc Herniations

The distinguishing histologic feature of recurrent disc herniation is the presence of large collagen bundles associated with a fibrillar framework.¹²⁸ Granulation tissue found in recurrences is not present in primary disc herniations. This fact indicates that the pathophysiology of recurrent disc herniations may be different than in primary cases.

Depending on the definition, recurrent disc herniations can occur in 18% of cases. Clinical presentation is usually of recurrent sciatic leg pain. Jonsson and Stromqvist¹²⁹ attempted to determine the relative frequency of clinical signs and symptoms after discectomy to differentiate better perineural fibrosis from true recurrent herniation. Pain reproduced by cough and a positive SLR were more frequent with recurrent disc herniations. These findings were also present in many

patients with fibrosis alone, however; thus, imaging modalities are crucial to the diagnosis. As discussed previously, MRI is best delayed for at least 6 months after surgery.

Recurrences can occur at the same level, same side, opposite side, or entirely different level. To be considered a recurrence, there must have been a pain-free (or relief) time period after the index surgery. A nonoperative treatment regimen, including a period of observation, physical therapy, medications, and other modalities, should be used in the initial treatment of recurrent disc herniations unless progressive neurologic deterioration occurs.

Important questions arise after the decision to proceed with surgery is made. The main focus of current investigations is whether or not comparable or inferior results can be expected after discectomy for a recurrent disc herniation. Some authors have documented inferior results after repeat surgery,¹³⁰ although most failures were reported in patients operated on without imaging evidence of neural compression.

Suk and colleagues¹³¹ retrospectively examined their results in a highly select group of patients. Recurrence was defined as a reherniation at the same level (ipsilateral or contralateral) after a pain-free interval of 6 months or more. These authors used gadolinium-enhanced MRI to confirm the diagnosis. They found that the second surgery was significantly longer and that recurrent disc herniations were typically larger. Clinical success was documented in 71.1% of cases, which was comparable to their results after primary discectomy (79.3%). In a similar study, Cinotti and colleagues¹³² examined their results with same-level contralateral reherniations only. At 2-year follow-up, surgery for recurrences resulted in satisfactory outcomes in 88% of cases and in 90% of primary discectomies. The only difference noted was more back pain in the reherniation group at 6 months, although this difference was not noted at 2 years.

Cinotti and colleagues¹³³ also studied a group of patients with ipsilateral lumbar disc reherniation and compared these patients with patients after primary surgery as a control group. Satisfactory results were reported in 85% of patients with ipsilateral lumbar disc reherniation and 88% of control patients. Noted differences were a higher degree of disc degeneration in the recurrence group. Although epidural fibrosis was abundant in the study group, its presence or amount did not adversely influence outcome. There was no difference in the psychological profiles of recurrent versus primary discectomy groups. In a similar study, Papadopoulos and colleagues¹³⁴ reported 85% improvement after discectomy for reherniations compared with 80% improvement for primary herniations at an average of 53.6 months follow-up. Herron¹³⁵ reported 69% good and 24% fair results after laminectomy and discectomy for recurrences at the same and other levels.

Most authors do not recommend fusion after a first-time recurrent disc herniation.^{135,136} After any lumbar surgery involving facetectomy and bone resection, preoperative and intraoperative assessment of stability is a major determinant of the decision to fuse. Although investigational evidence is lacking, we consider fusion after a second reherniation at the same level.

Discectomy in Children

The role of trauma in causing disc herniations in children is believed to be more significant than in adults.^{3,137,138} Children are unlikely to have disc degeneration or the antecedent period of back pain before herniation. Clinical signs and symptoms are similar to adults but are much more acute in onset. The clinical and imaging evaluation is similar to adults.

The literature of discectomy in children reveals prolonged length of follow-up. Initial results are excellent, but results tend to deteriorate with time. At 1-year follow-up, Papagenlopoulos and colleagues¹³⁹ reported 93% good or excellent results. At final follow-up, ranging from 12 to 45 years, 92% were good or excellent; however, there was a 28% reoperation rate. Parisini and colleagues¹⁴⁰ documented 95% success at short-term follow-up, whereas at long-term follow-up (average, 12.4 years), good or excellent outcomes were documented in 87% of cases. These investigators found a 10% reoperation rate at 10 years. DeOrion and Bianco¹⁴¹ showed 96% good or excellent results at initial follow-up, trending down to 74% at final examination. Numerous other series document similar results.^{3,137,142}

Fusion has been reported in a small percentage of pediatric patients. Well-defined selection criteria are lacking, however. The influence of congenital anomalies has been considered previously. There does not seem to be a correlation between any one anomaly and disc herniation. It is important to recognize anomalies, such as spina bifida, during routine preoperative planning to avoid iatrogenic neurologic injury.

Discectomy in Elderly Patients

Lumbar disc herniations are much less common in elderly patients. The nucleus desiccates with age and is less likely to herniate. Underlying stenosis and bony overgrowth is a more common problem. Regardless, herniations can still occur, with or without the presence of stenosis. The results of discectomy in this population are comparable to younger patients, assuming the correct diagnosis is made.

Maistrelli and colleagues¹⁴³ reported results of discectomy in 32 patients older than 60 years. Clinical findings were similar, with 81% of patients having root tension signs such as a positive SLR. At an average follow-up of 50 months, good or excellent results were found in 87% of cases. None of the patients had evidence of neurogenic claudication, which is an important distinguishing feature when making the diagnosis of a disc herniation versus stenosis. Jonsson and Stromqvist¹⁴⁴ also found disc surgery to be gratifying in patients older than 70 years, with good results documented at 2-year follow-up.

Foraminal and Extraforaminal (Far-Lateral) Herniations

So-called far-lateral disc herniations require special consideration. Surgical excision is more easily accomplished through

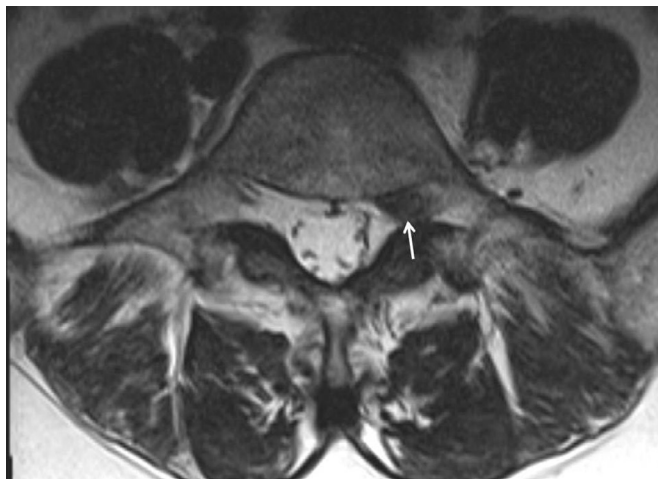


FIG. 47.16 Magnetic resonance imaging can readily show foraminal disc herniations that may otherwise be missed by computed tomographic or plain myelograms because neural compression is extradural.

modified operative techniques. Foraminal and extraforaminal disc herniations are more common in older patients. Symptoms are more likely to be isolated to one particular nerve root. In contrast to paracentral discs, they lead to compression of the exiting nerve root rather than the descending root. Diagnosis is best by MRI or CT (Fig. 47.16). Because the compression is usually extradural, myelography does not show a filling defect.

Different techniques have been advocated for the foraminal or extraforaminal herniation. Compression of the nerve root occurs outside of the spinal canal, making visualization of the herniation challenging through a standard laminotomy or laminectomy performed from within the canal. Garrido and Connaughton¹⁴⁵ advocated a unilateral facetectomy. Bone is removed along the entire path of the nerve root from its exit from the cauda equina out and through the foramen; this necessitates complete removal of the facet on the side of exposure. Although symptom relief was not well documented, painful radiographic instability was detected in only 1 of 35 (3%) cases. This patient had pain relief after fusion.

Donaldson and colleagues¹⁴⁶ advocated a less destabilizing procedure. They considered identification of the nerve within the intertransverse process “blind.” As a solution, they performed a partial hemilaminectomy at the level above the disc at the point where the root exited the cauda equina. They followed it laterally beneath the bone of the remaining pars interarticularis. By placing a probe over the root and advancing it laterally, they were able to identify the root and facilitate its dissection between the transverse processes before discectomy was performed through this interval. They reported good or excellent pain relief in 72% of patients. No cases of instability were reported.

Melvill and Baxter¹⁴ used the intertransverse approach for discectomy in 40 patients with extraforaminal herniations. They performed this using standard midline dissection that was extended subperiosteally out and over the facets to expose the transverse processes. This is in contrast to the classic description of the Wiltse paraspinal muscle-splitting approach.

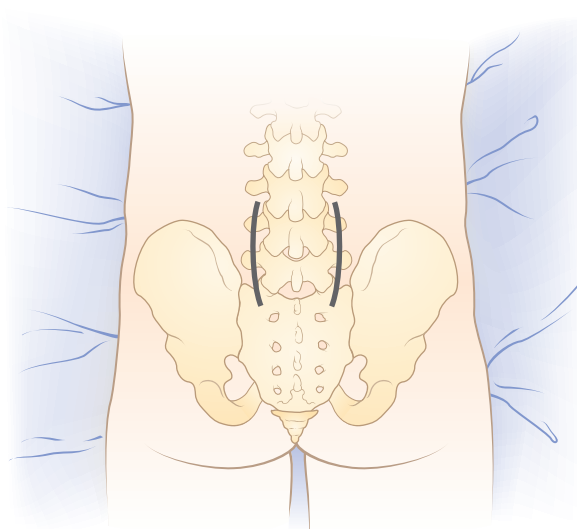


FIG. 47.17 The Wiltse muscle-splitting approach, using a paramedian incision, can be used to access the intertransverse process region.

Complete resolution of leg pain was reported in 85% of cases. Melvill and Baxter¹⁴ noted a fibrous band that was present tethering the nerve root to the lateral aspect of the disc in an associated cadaveric study.

In an investigation with midterm follow-up, satisfactory results were reported using the Wiltse paraspinal muscle-splitting approach for so-called lateral disc herniations.¹⁴⁷ Satisfaction was reported by 85% of patients, with 60% of patients reporting complete resolution of pain. At 5 years' follow-up, 20% of patients had developed some degree of instability, half of whom ultimately underwent fusion. Assuming uniformity of the definition of instability and indications for fusion in these cases, this finding is supportive of the hypothesis by McCulloch and Transfelt³¹ that lateral disc herniations were a precursor to the development of degenerative spondylolisthesis.

A thorough knowledge of the anatomy in this region is imperative to performing an effective discectomy without undue bleeding or injury to the nerve root.⁸⁹ We prefer to excise foraminal or extraforaminal discs through a true Wiltse paraspinal muscle-splitting approach to the intertransverse interval. It is easiest through a paramedian incision (Fig. 47.17). Identification of the correct level of incision is crucial in minimizing the length of the incision.

Cauda Equina Syndrome

Cauda equina syndrome (CES) most commonly occurs from a herniated lumbar disc. It is more common with central herniation (27% of central herniations), although it can occur with paracentral or lateral herniations as well.^{148,149} It is more frequent in men in their fourth decade. The odds of developing CES are significantly higher in overweight and obese individuals.¹⁵⁰ An L4–L5 disc is the usual cause.¹⁴⁸ CES should be considered a true surgical urgency because neurologic results are affected by the time to decompression. The clinical

Pain:
Backs of thighs and legs

Numbness:
Buttocks, backs of legs,
soles of feet

Weakness:
Paralysis of legs and feet

Atrophy:
Calves

Paralysis:
Bladder and bowel

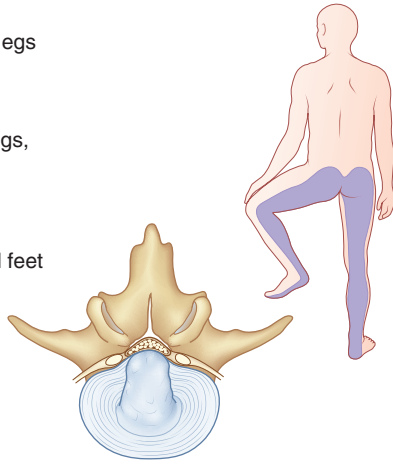


FIG. 47.18 Cauda equina syndrome is characterized by saddle anesthesia, motor weakness, and loss of bowel and bladder control.

diagnosis of CES relies on many components, including perineal sensory deficit (so-called saddle anesthesia), bowel or bladder incontinence, new-onset lower extremity sensory deficit, and a new or progressive motor deficit (Fig. 47.18). In addition to a meticulous physical examination, evaluation of CES should include measurement of a bladder postvoid residual. Normally, the postvoid residual should be less than 50 to 100 mL. The postvoid residual is often abnormal preoperatively and can be an important parameter to follow postoperatively.¹⁵¹

Decompression and discectomy can be via a laminotomy or through a formal laminectomy. Proponents of laminectomy believe that this provides superior visualization of the dura and avoids excessive traction.^{152–154} Adequate exposure is particularly relevant for removal of central disc herniations.

The effect of timing of surgery on outcomes continues to be debated. Although a number of studies have reported that surgical decompression within 48 hours of the onset of symptoms is a safe time point within which to perform surgery, more recent systematic reviews have concluded that there is no strong support for the often-cited 24- or 48-hour thresholds nor that the earlier the decompression, the better the outcomes.^{151,154–156}

DeLong et al., in a metaanalysis of a number of observational cohort series, found a statistically significant relative risk for timing of surgery greater than 24 hours.¹⁵⁷ These authors remarked that higher-quality scientific study is difficult to obtain on this subject since equipoise is lacking, making a prospective randomized study difficult to execute. In analyzing the effect of early surgical decompression on urinary function, it is important to differentiate patients who have incomplete cauda equina syndrome (CES), characterized as “altered urinary sensation, loss of desire to void, poor urinary stream, and the need to strain in order to micturate” from CES with retention, characterized as “painless urinary retention and overflow incontinence, when the bladder is no longer under executive control,” the former of which was found to have a better prognosis.¹⁵⁸ In one study, decompressive surgery within 24 hours of onset of autonomic symptoms in CES

reduces bladder dysfunction at initial follow-up, while no statistically significant difference was observed in CES with retention regarding timing of operation.¹⁵⁹

Strength may continue to improve for 1 year after surgery.¹⁵⁴ Although the postvoid residual usually decreases to less than 110 mL by 6 weeks, bladder function may continue to improve for 16 months.¹⁴⁹ Early surgery does not seem to substantially affect the resolution of postoperative pain compared with delayed intervention.¹⁶⁰ In a systematic review of the literature reporting dysfunction of micturition, defecation, and/or sexual dysfunction following CES secondary to herniated disc, Korse and colleagues reported considerable continued impairment. At a minimum of 17 months follow-up, 42.5% of patients reported micturition dysfunction, 49.6% reported defecation dysfunction, and 44% reported sexual dysfunction.¹⁶¹ Above all, preoperative neurologic status seems to be the greatest predictor of recovery.¹⁶² This is evident in that many patients are found to have residual neurologic deficits despite timely and effective surgical intervention.¹⁶⁰

Alternatives to Standard Discectomy

Endoscopic Discectomy

The use of the endoscope to perform lumbar discectomy has enthusiastic proponents. Improvements in equipment have made the technique more user-friendly and facile. Still, endoscopic discectomy remains a technically demanding procedure, and the results depend heavily on surgeon experience.¹⁶³ It is beneficial to have prior training in minimally invasive spine surgery.¹⁶⁴ Although it is not yet a replacement for standard discectomy techniques, more recent series have shown the ability to achieve comparable results in some surgeons' hands.^{80,165,166}

Purported advantages of endoscopic lumbar discectomy include less intraoperative blood loss and smaller surgical wounds. A study by Pan and colleagues did not show a significant difference in patient-reported pain when comparing endoscopic and traditional open discectomy, though postoperative inflammatory markers such as interleukin-6 and C-reactive protein were reduced in the endoscopic group.¹⁶⁷ DeAntoni and colleagues⁹⁶ reported results of translaminar epidural endoscopic discectomy in 190 patients. Using a lateral decubitus position, the technique uses a small paramedian incision for introduction of the instruments. Dilators are used to strip the muscles subperiosteally from the adjacent lamina as far lateral as the facet joint. The arthroscope (endoscope) is inserted through a 6-mm working cannula. An additional lateral paraspinal incision is made for an outflow cannula. A shaver is used to remove bone within the interlaminar window until the attachments of the ligamentum flavum can be visualized. The ligamentum flavum is elevated using a Penfield dissector and removed with a Kerrison rongeur, similar to a standard open approach. A root retractor pulls the nerve root and dura medially, allowing access to the disc space. The annulus is incised, and the herniation is removed. The endoscope can be inserted into the disc space to look for any additional loose fragments (so-called discoscopy). Good or

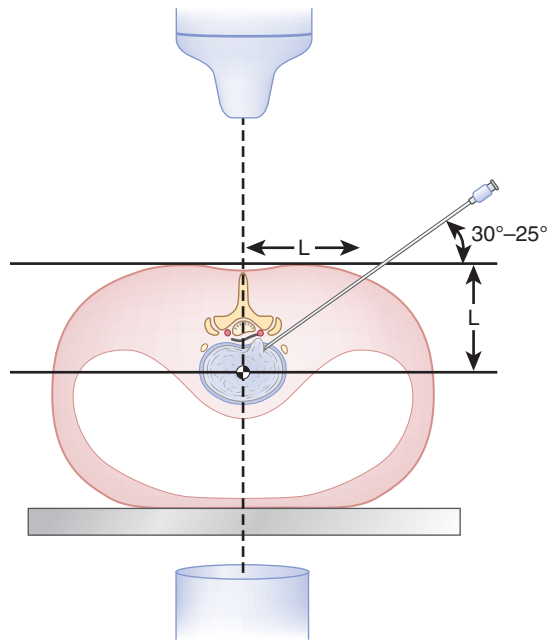


FIG. 47.19 The transforaminal endoscopic approach for disc excision. (From Yeung AT, Tsou PM. Posterolateral endoscopic excision for lumbar disc herniation. *Spine (Phila Pa 1976)*. 2002;27:722–731.)

excellent results were documented in 92% of cases with a minimum 2-year follow-up. There was only one dural laceration that did not need repair, there was no instability, and there were no neurologic injuries.

Yeung and Tsou⁸⁰ reported results of 307 endoscopically assisted transforaminal discectomies. The patient is positioned supine on a radiolucent table. Local anesthesia with light sedation is used. In this technique, intraoperative fluoroscopy is crucial. A standard discogram is performed first, injecting blue dye into the disc space to help identify the herniated tissue. Along the same path, a series of dilators and ultimately a single working cannula is inserted percutaneously through the intervertebral foramen (Fig. 47.19). One cannula is used for inflow, outflow, and instrument insertion. The cannula has a shielded tip that allows aggressive discectomy with the use of graspers and shavers while retracting the exiting nerve root. As stated, this is performed through the intervertebral foramen. The nerve root can be visualized by rotating the cannula, allowing assessment of its integrity and mobility. Satisfactory results were reported in 89.3% of patients. Six patients (2%) had lower extremity dysesthesia that lasted longer than 6 weeks, two had thrombophlebitis, two had discitis, and one had a dural tear that did not require repair. Yoshimoto and colleagues demonstrated transforaminal microendoscopic decompression to be effective for treating far-lateral disc herniations.¹⁶⁸

This technique has been shown to be safe and effective for recurrent disc herniations as well.¹⁶⁶ In a prospective evaluation of 262 consecutive cases of reherniation treated by an endoscopically assisted transforaminal technique, Hoogland and colleagues¹⁶⁶ reported an 85% success rate and 3.8% complication rate at 2-year follow-up. Kim and colleagues reported that percutaneous endoscopic interlaminar discec-

tomy can be successful for reherniation by avoiding the iliac crest, which can be prohibitive with transforaminal techniques at L4–L5 and L5–S1.¹⁶⁹

Lee and colleagues prospectively studied the influence of patient position on the location of abdominal organs using preoperative prone and supine CT scans, and found that safe approach angles were significantly more horizontal in the prone position.¹⁷⁰ The prone abdominal CT scan was more helpful in determining the trajectory and feasibility of the endoscope. A true lateral approach was possible for 75% of the discs at L3–L4 and 76.7% of the discs at L4–L5.

Several authors have sought to compare the results of microendoscopic discectomy with more traditional techniques.^{171–173} Ruetten and colleagues¹⁷³ compared the results of endoscopic interlaminar and transforaminal discectomy with microsurgical discectomy. They showed no clinically significant differences between the endoscopic and microsurgical groups and identical reherniation rates (6.2% for each).¹⁷³ Analogous findings have been reported with comparisons of endoscopic and standard open discectomy.^{171,172} However, Teli et al. reported higher rates of dural tears and recurrent herniation in endoscopic surgery compared to traditional techniques, with similar patient-reported outcomes at 2 years.¹⁷⁴ Garg et al., in a prospective randomized study, reported shorter hospital stay and earlier return to work in the microendoscopic group compared to the open discectomy group.^{174a} In a systematic review of randomized controlled studies comparing microendoscopic technique with open discectomy or microdiscectomy, Smith and colleagues found that patient-reported outcome measures were similar with all three techniques.¹⁷⁵

In a recent retrospective review of over 10,000 patients who had undergone percutaneous endoscopic lumbar discectomy, Choi and colleagues sought to find common causes of surgical failure, defined as a case requiring reoperation within 6 weeks.¹⁷⁶ They found that 4.3% of cases were failures, and over half of these were due to incomplete removal of herniated disc, of which inappropriate positioning of the working channel was the cause in 33.6% of cases. Exiting root injury is another complication of endoscopic surgery. Choi et al. analyzed the radiologic risk factors for exiting root injury by retrospectively reviewing the preoperative MRIs of patients undergoing endoscopic discectomy who experienced this complication.¹⁷⁷ They determined that if the distance between the exiting nerve root and the superior articular process of the caudal level is inadequate to accept the diameter of the working cannula without compressing the nerve root, an alternative technique, such as microdiscectomy or open discectomy, should be employed.

Percutaneous Automated Discectomy

Because of the evolution of percutaneous techniques of discectomy, there is considerable “overlap” in the names of different procedures. Although endoscopic discectomy can be considered percutaneous, it is not what is typically considered *percutaneous automated discectomy*. This term refers specifically to a procedure introduced in the early 1980s.

Using a posterolateral approach, a tissue-removing device is introduced into the disc space in a similar path as the needle of a discogram; this is guided by intraoperative fluoroscopy. Earlier techniques used a direct lateral approach that led to (or raised concern of) viscous perforation in some patients. The instrument is used to remove nucleus material in small increments. This removal theoretically can decrease intradiscal pressure. There is no direct visualization of the nerve roots, dura, or disc material. Neurologic decompression cannot be assessed with this technique.

Results of automated discectomy are inferior to standard open discectomy. Shapiro⁹³ showed only partial improvement of leg pain in 57% of patients undergoing this procedure. More disappointing was that only 5% had complete sciatica relief. All patients had disc bulges or protrusions, with none having free fragments. These results are less positive than the results documented with open discectomy.

In contrast, Hoppenfeld⁹⁴ reported more successful results, with relief of sciatica and sensory deficit in 86% of patients. He determined, however, that patients with sequestered discs do not reliably respond well to the procedure. Kotilainen and Valtonen⁹⁵ treated 41 patients with small protrusions or prolapses (bulges) with percutaneous automated discectomy. Sciatica had completely resolved in 78% of patients. From these data, it can be inferred that automated discectomy can be reasonably effective in patients with small disc bulges or protrusions that are in direct continuity with the remaining nucleus. Only in this select group of patients is there the possibility of relieving the neural compression. The procedure should not be used in patients with sequestered or extruded fragments (uncontained herniations) or most patients with disc herniations.

Chemonucleolysis

Chemonucleolysis involves the chemical digestion of nucleus material via injection of an agent, such as chymopapain, into the intervertebral disc. Ideally, the agent not only decompresses the central aspect of the disc space (producing a similar effect as percutaneous automated discectomy), but also might directly attack the herniated fragment. The clinical success of the procedure has not been supportive.

Chemonucleolysis has been rigorously examined in numerous clinical investigations. In a randomized prospective trial comparing chemonucleolysis and standard discectomy, Muralikuttan and colleagues⁹⁸ concluded that it leads to inferior results. There was a high rate of conversion to open surgery in the chemonucleolysis group because of continued, unrelieved symptoms. Crawshaw and colleagues¹⁷⁸ found extremely high failure rates—47% to 52%—after chemonucleolysis compared with 11% with open surgery. van Alphen and colleagues⁸² documented increased radicular pain after chemonucleolysis in 22% of patients treated. Salvage discectomy after failed chemonucleolysis leads to worse results than primary surgery.^{82,98,178} In addition to these poor results, neurologic complications, such as transverse myelitis associated with the use of chymopapain, have all but eliminated the procedure from continued use in the United States.

PEARLS

1. Most lumbar disc herniations respond well to conservative treatment within the first 3 months from the onset of symptoms.
2. In patients who fail conservative treatment, surgery consistently showed better outcomes than continued nonoperative care.
3. The type and size of disc herniations and psychosocial factors are the primary determinants of outcomes after discectomy.
4. Although minimally invasive techniques, such as endoscopic and percutaneous laser discectomy, can be effective in experienced hands, it is unclear whether they would ever replace the gold standard operation of open discectomy (with or without the use of a microscope).

PITFALLS

1. Patients with extraforaminal (far lateral) and central disc herniations should be advised that surgical outcomes may be inferior compared with outcomes for more common and typical paracentral herniations.
2. Patients should be informed that a lumbar discectomy is primarily indicated for leg pain. Back pain, numbness, and weakness are less reliably alleviated after surgery.
3. Recurrent disc herniations, which can occur in 18% of patients, remain problematic and may be more prevalent in patients who have large annular defects after discectomy.
4. Although simple discectomy for a recurrent disc herniation can yield outcomes equivalent to the index procedure, the addition of fusion may be considered after a second recurrence.

KEY POINTS

1. Patients should be carefully counseled preoperatively on the risk and benefits of surgery so that they may make a well-informed decision according to their specific functional demands and expectations.
2. Positioning a patient in a flexed or kneeling position can help open the interlaminar window to allow easier entry into the spinal canal.
3. Regardless of the technique of discectomy, a successful operation is contingent on adequate exposure that enables careful and minimally traumatic identification of the cauda equina, descending nerve root, and exiting nerve root before retraction to retrieve the herniated fragments.
4. During discectomy, the amount and location of the disc herniation should be commensurate with that shown on preoperative MRI.
5. The spinal canal and disc space should be thoroughly inspected before closure to avoid retained disc fragments.

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